



MANAGEMENT OF PSYCHIATRIC DISORDERS



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RANZCP EXAM PREP



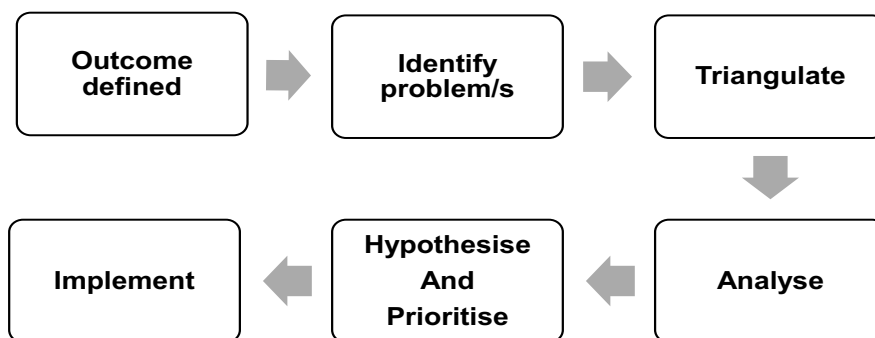
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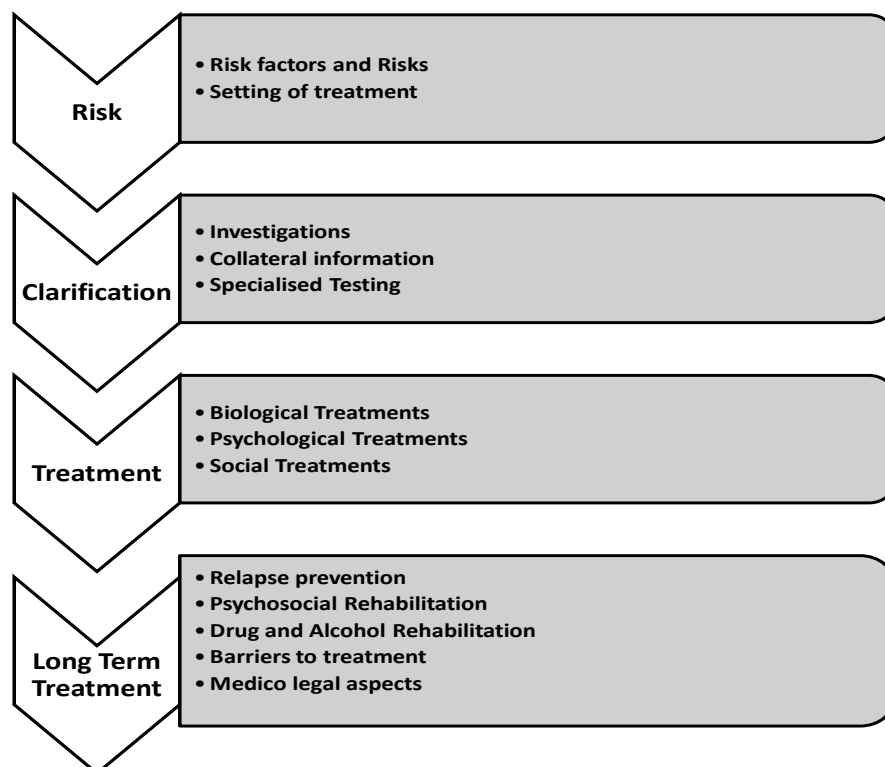
MANAGEMENT OF PSYCHIATRIC DISORDERS

MANAGEMENT OF MENTAL DISORDERS

Strategic Problem Solving



DOMAINS IN MANAGEMENT OF PSYCHIATRIC DISORDERS

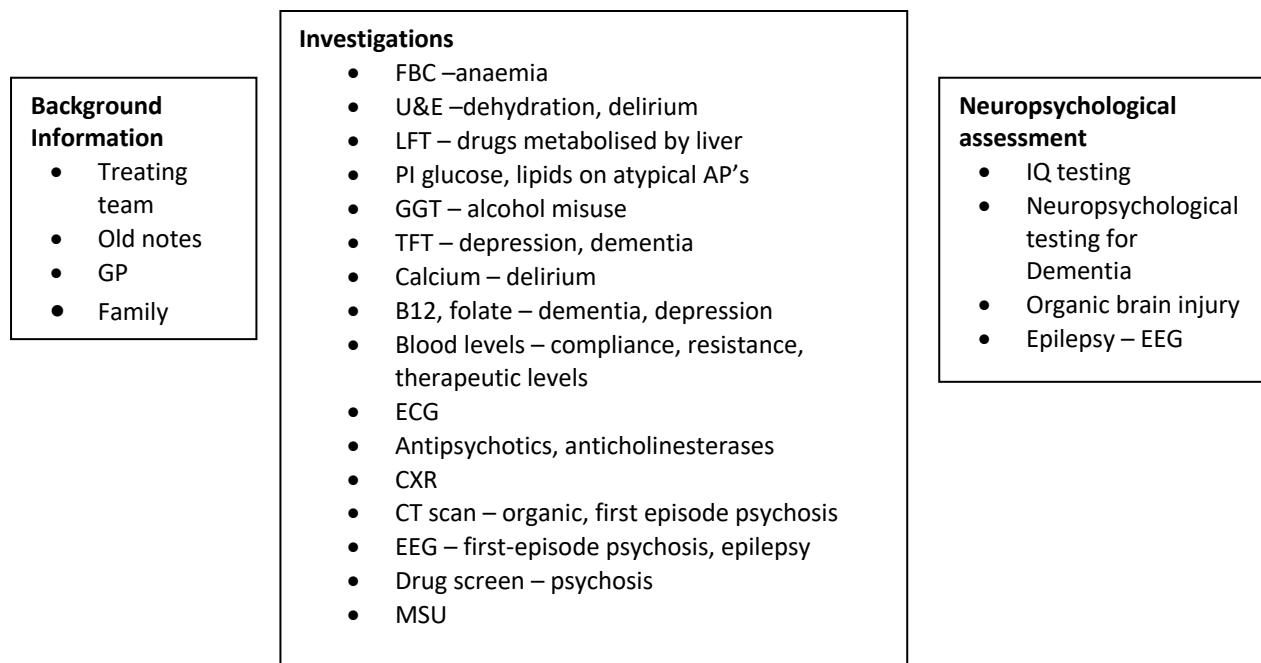


1. RISK

This consists of the following domains

- **Physical risk** – This can be due to medical problems, e.g. diabetes, starvation in anorexia or other medical conditions.
- **Risk of suicide or self-harm.**
- **Risk of homicide or harm to others** –
 - This is especially relevant in a forensic case and, as mentioned in the previous chapter, should include a risk formulation or in case of time not being available, be presented in this section.
 - The risk formulation can be based on HCR-20 providing both mitigating and aggravating factors
- **Risk to reputation**
- **Risk to children** – neglect (drug and alcohol abuse by parents)
- **Risk to finances** – mania, overspending, gambling, debt
- **Driving**
- **Corporate Risk** – professionals, persons in positions of responsibility (e.g., doctors, nurses, and accountants) may prove a risk to employer organisation.
- **Risk of self-neglect**
- **Risk of non-compliance/relapse**
- **Risk of substance misuse**

2. CLARIFICATION OF DIAGNOSIS



Scales for **Dementia:** ADAS-Cog
Alcohol: CIWA-R
Schizophrenia: PANSS, BPRS
Depression: HAM-D / MADRS
Mania: YMRS

3. TREATMENT OF ACUTE PSYCHOLOGICAL SYMPTOMS/MAINTENANCE OF EUTHYMIA/STATUS QUO (ESP. IN OUT-PATIENTS)

- Medication – depending on condition
- Nursing support – observation
- Education
- Groups, etc.

4. LONG-TERM MANAGEMENT

Compliance

- Discuss reasons for non-compliance
- Increasing compliance
- Techniques – Motivational interviewing
 - Compliance therapy
 - Cognitive techniques

Maintaining abstinence from drug and alcohol

Psychosocial rehabilitation/rehabilitation and recovery

- Accommodation– high level or low-level supported accommodation
- Supported employment/pre-vocational training
- Education
- Finances – carers allowance, benefits
- Groups, day services/hospitals

Medico- Legal issues

PSYCHIATRIC DISORDERS

The techniques for presenting management plans is discussed during the course. Following are the most relevant ‘Need to Know’ points for psychiatric disorders. The aim is to build a good fundamental knowledge base and to incorporate the following in a structured management plan (discussed in detail and practised during the course)

ANXIETY DISORDERS

GENERALISED ANXIETY DISORDER

Generalised anxiety disorder (GAD) is characterised by persistent feelings of fear and worry about everyday things that are difficult to control [APA 2013].

Epidemiological research shows that:

- The median age of onset is in the early 30’s and it is two times more prevalent in women than in men.
- The general prevalence in children under the age of 18 is between 5.7% and 12.8% and there is an estimated lifetime prevalence of 6%.
- GAD is more common in individuals that are separated, divorced, or widowed.
- Low educational attainment and unemployment are also consistently associated with GAD, which may be related to an individual’s poor level of cognitive skills i.e. memory and problem solving.

DIAGNOSIS

GAD can involve excessive worrying about singular or multifocal things such as finances, family, health, work and/or study, community, and world affairs. GAD is diagnosed if three of the following 6-items have been present for more than 6 months:

- Restlessness
- Easily fatigued
- Concentration problems
- Muscle tension
- Sleep disturbance issues
- Irritability

GAD is on average only diagnosed 10 years after onset at which point the patient has gone to the doctor’s due to comorbid issues such as panic disorder, depression, or chronic pain disorders [Andrews et al 2018]. As such, because of the high comorbidity of anxiety and depression, all patients that are diagnosed with anxiety should be screened for depression as well.

INITIAL ASSESSMENT AND MANAGEMENT

Initial assessment should address behaviour and somatic symptoms; the evaluation of psychosocial stress and developmental issues in the context of past medical history can be used to exclude other organic causes.

Tests for thyroid function, blood glucose level, echocardiography, and a toxicology screen are important at this early stage.

The brief **GAD-7 self-report assessment** is a validated diagnostic tool that detects a patient's latent anxiety and how this affects their ability to respond to an item in a particular manner. There are 7 questions and the responses can be either: "not at all", "several days", "over half the days", and "nearly every day" [Spitzer et al 2006]:

1. Feeling nervous, anxious or on edge.
2. Not being able to stop or control worrying.
3. Worrying too much about different things.
4. Trouble relaxing.
5. Being so restless that it is hard to sit still.
6. Becoming easily annoyed or irritable.
7. Feeling afraid as if something awful might happen.

Alternatively, the **Penn State Worry Questionnaire-3 (PSWQ-3)** is an ultra-brief 3-item version of the PSWQ, which has 16-items [Meyer et al 1990; Berle et al 2011]. The 3-item version is suggested to have similar psychometric properties as the full PSWG version and has been confirmed to detect a good measure of pathological worry:

- Many situations make me worry.
- Once I start worrying, I cannot stop.
- I worry all the time.

In youth populations, the **Paediatric Anxiety Rating Scale (PARS)** has been used in multiple psychopharmacological studies involving youth with GAD [Birmaher et al 2003; Walkup et al 2008; Strawn et al 2015]. PARS includes a 50-item checklist that encompasses anxiety related to social, performance, separation, generalised, specific phobias, as well as somatic symptoms.

CLINICAL RECOMMENDATIONS

As with all disorders across the anxiety spectrum, a pragmatic approach that includes psychoeducation and information on lifestyle factors (e.g. healthy lifestyle choices concerning diet, exercise, and sleep) is encouraged [Andrews et al 2018].

CBT is considered more effective than other psychotherapies and should include 8 to 12 sessions, although interestingly The Cochrane Database of Systematic Reviews suggest that 5 to 8 sessions may produce more benefits in GAD patients than ≥ 9 sessions of CBT [Hunot et al 2007].

TREATMENT

Delivery of CBT (face-to-face or digitally) is the first-choice option as there is evidence that both forms of CBT are more effective than pharmacotherapy [Cuijpers et al 2014]. However, this is also dependent on severity, as well as patient preference, accessibility, cost, and tolerability. CBT has three stages:

1. First stage is structured to include **education about worrying** and the factors that facilitate or hinder therapy. Cardio exercise is encouraged three times per week and the topic of misuse of alcohol and drugs is broached.
2. **Progressive muscle relaxation and breathing training** to improve physical symptoms as well as reducing cognitive symptoms to challenge unhelpful thinking. In addition, patients are confronted with behaviour avoidance using graded exposure.
3. The final stage is aimed at **relapse prevention** so that any benefits are maintained at 3-months, 1-year, or 3-years post-treatment. Identification of early warning signs is an important feature as well as future goal setting.

CBT should be started as early as possible, and if this is not feasible then digital CBT should be initiated [Robinson et al 2010].

CBT should be individual rather than group CBT as there is earlier improvement and greater reductions in worry and depressive symptoms as well as higher adherence rates (9% vs. 24% attrition rates, respectively) [Hunot et al 2007; Covin et al 2008].

Other psychotherapy alternatives such as applied relaxation, mindfulness, and acceptance-based interventions have limited evidence.

SSRIs (e.g. sertraline, paroxetine, and escitalopram) and SNRIs (e.g. venlafaxine and duloxetine) are supported by RCT data to specifically reduce symptoms and remission rates, respectively [Andrews et al 2018].

MONITORING PROGRESS

After 4 to 6 weeks of either CBT or medication, the response should be reviewed for efficacy and safety [Andrews et al 2018]. If there is a partial efficacy, then treatment continues. However, if not, then modifications can be made based on what treatment strategy was initially decided.

- If dCBT was selected, then face-to-face CBT should be attempted; if face-to-face CBT was selected, then add a SSRI; if pharmacotherapy was selected, then add CBT or increase drug dosage.
- If these modifications are unsuccessful after a further 4 weeks, then combination of CBT and SSRI should be attempted if not already done. CBT session frequency can be increased but a second opinion and another review of the diagnosis and comorbidities may be necessary.

- Pregabalin and short-term agomelatine treatment have also shown to have a degree of efficacy and should be considered if there hasn't been a response to a SSRI/SNRI. However, agomelatine is not approved for the treatment of GAD in Australia.
- An alternative approach is with the use of benzodiazepines but only for **short-term regular use**. However, there are significant side effects and as such are generally not recommended for the treatment of GAD in primary or secondary care.

SUMMARY

Clinicians should be aware of the myriad of comorbid disorders that often present with GAD. Evidence suggests psychoeducation and lifestyle changes as well as CBT as the most effective forms of therapy.

PANIC DISORDER

Panic attacks are characterised as a discrete period of intense fear that develops abruptly and reaches a peak within 10 minutes.

Panic disorder is defined as recurrent periods of panic attacks whereby individuals are persistently concerned about when the next panic attack will start and what the consequences will be if they lose control.

DIAGNOSIS

Diagnostic interviews have consistently reported symptoms such as heart pounding (97%) and dizziness (96%), however there are a number of other reported symptoms [Cox et al 1994]. However, these clinical presentations are not generic specifiers as they can also be substance-induced (e.g. stimulant or caffeine intoxication or withdrawal from alcohol or benzodiazepines) or identified in other disorders.

SYMPTOMS – Four of the following symptoms:

- Palpitations
- shortness of breath
- feelings of choking
- chest pain
- nausea
- light-headedness
- derealisation/depersonalisation
- losing control
- fear of dying
- paraesthesias
- hot or cold flushing.

- Type – Panic attacks can be unexpected, or they can be expected whereby there is anticipatory anxiety that a panic attack is associated with immediate physical (e.g. heart attack), mental (e.g. loss of control), or social (e.g. embarrassment) symptoms.
- Situation – Behaviours are changed to avoid situations such as being in a crowd (or being alone) or it can be a fear of being in an enclosed space (or in an open space).

Separate diagnosis of panic disorder with agoraphobia is also included in the DSM-5 as it has been reported that up to **50% of patients with panic disorder also have agoraphobia** [Goodwin et al 2005; Bienvenu et al 2006].

Agoraphobia is the fear of being caught in a situation where escape to a safe place may be difficult—and as such many patients will change behaviours to avoid these situations.

Marked fear or anxiety about two (or more) of the following five situations

1. Using public transportation
 2. Being in open spaces
 3. Being in enclosed spaces
 4. Standing in line or being in a crowd
 5. Being outside of the home alone
- The individual fears or avoids these situations because of thoughts that escape might be difficult or help might not be available in the event of developing panic-like symptoms or embarrassing symptoms
 - The agoraphobic situations almost always provoke fear or anxiety
 - The agoraphobic situations are actively avoided, require the presence of a companion or are endured with intense fear or anxiety
 - The fear or danger is out of proportion to the actual threat
 - Duration of 6 months or more
 - Clinically significant distress or impairment in social, occupational or other important areas of functioning

INITIAL ASSESSMENT AND MANAGEMENT

After establishing a good relationship, the clinician should determine

1. The primary diagnosis is panic disorder rather than any other anxiety-related diagnosis
2. Whether agoraphobia is present or not, and
3. The identification of any co-morbid anxiety, mood disorders, or substance abuse as this will aid in understanding treatment efficacy as well as the long-term prognosis.

Furthermore, the clinician should investigate whether the patient has any biopsychosocial or lifestyle factors that trigger panic attacks or indeed whether any of these factors were what precipitated or are maintain the disorder. To obtain this information, a comprehensive assessment should cover the following points [Andrews et al 2018]:

- Severity and duration of symptoms and has the patient received prior treatment?
- Level of distress or functional impairment is the patient experiencing and is the patient a suicide risk?

- Does the patient have any co-morbid psychiatric disorders or chronic health issues?
- Mental health history of the patient and family and is there evidence of domestic violence or sexual abuse?
- Quality of the patient's interpersonal relationships and support network and are these affected by employment status, social isolation, living conditions, immigration status?
- Full medical evaluation including blood tests and physical examination with relevant family history also explored.
- Goals and expectations of the patient?

CLINICAL RECOMMENDATIONS

TREATMENT

Psychoeducation on lifestyle factors (i.e. improved eating, sleeping, and exercising) followed by **CBT** with a clinical psychologist is supported by several studies to reduce panic attacks [Andrews et al 2018].

CBT can be face-to-face and delivered over 8 to 12 sessions or it can be provided as guided digital CBT (dCBT):

- Stage 1 – Provides psychoeducation on anxiety and panic disorder as well as the rationale for treatment. Factors that may facilitate successful therapy should be identified, which may include the future education of family and social support members.
- Stage 2 – Includes behavioural experiments that can challenge a patient to identify catastrophic cognitions that hinder therapy. Patients are required to identify and reduce feared physical sensations and avoidance behaviours that the patient uses to prevent panic-related situations.
- Stage 3 – Structured to prevent relapse by identifying what may precipitate a relapse and strategising how the patient can manage these situations.

First-line antidepressant pharmacotherapy with SSRIs or SNRIs have been shown to be effective across many studies [Mitte et al 2005; Andrisano et al 2013]. Although not a clinical recommendation, evidence suggests that efficacy is variable with paroxetine and fluoxetine exhibiting a degree of superiority over citalopram or venlafaxine.

Benzodiazepines (i.e. alprazolam) have also been shown to have a moderate effect size similar to SSRIs however these drugs are no longer first-line agents due to their side effect profile and dependency potential [Mitte et al 2005]. There is limited to no evidence available for anxiolytics, anticonvulsants, antipsychotics, beta-blockers, or for other antidepressants.

MONITORING PROGRESS

- Self-report questionnaires – **The Panic Disorder Severity Scale (PDSS)** and **Panic Agoraphobic Spectrum (PAS)** are available as interview instruments or self-report questionnaires (7-items and 114-items, respectively). These questionnaires measure the frequency and severity of panic attack symptoms as well as the severity of

anticipatory anxiety and phobic avoidance however the PAS also includes separation sensitivity, stress sensitivity, medication sensitivity, and agoraphobia domains.

- Guided assessments – The well-validated **14-item Agoraphobic Cognitions Questionnaire (ACQ)** has a 5-point scale that assesses the severity of cognitions, loss of control, and physical concerns. Alternatively, The Anxiety Sensitivity Index (which measures an individual's tendency to misinterpret anxiety-relevant cues) or The Mobile Inventory Avoidance scale (which measures agoraphobic avoidance) can also be used.

SUMMARY

Clinical practice guidelines recommendations suggest that treatment should be guided by the patient however **CBT is preferred as a first-line choice of treatment before pharmacological intervention**. This is assuming that this option is available, is not contradicted by any comorbidity, and that the patient has not had an absence of response to pharmacotherapy previously.

SOCIAL ANXIETY DISORDER

Social anxiety disorder (SAD) is an intense social discomfort of being negatively evaluated or judged by other people. Individuals believe that their fear is making them act in a way that other people will be offended by—as such their fear of social inappropriateness results in avoidance behaviours and impaired social functioning [APA 2013].

- Up to 8.4% of people are suggested to meet SAD criteria over a 12-month period.
- SAD is more common in women than in men (1.2- to 1.5-fold) and is more common in separated, divorced, or never-married individuals.
- Only 20 to 40% of patients diagnosed with SAD will recover within 20 years of onset [Ruscio et al 2008].

DIAGNOSIS

SAD was initially termed social phobia and is often misdiagnosed with agoraphobia due to the similarity in symptoms related to a fear of social situations (e.g. crowded spaces).

However, to differentiate SAD from agoraphobia requires an understanding of the underlying cause: negative evaluations of themselves in SAD versus fear of physical or mental harm in agoraphobia [APA 2013].

SAD has many comorbid diagnoses however because of its early onset, SAD is often the initial diagnosis.

Therefore, the DSM-5 diagnosis for SAD includes the presence of a high number of psychiatric and medical conditions, particularly within the spectrum of anxiety disorders, mood disorders, and substance abuse disorders [Andrews et al 2002].

Therefore, distinguishing SAD from other diagnoses is difficult:

- Symptoms of MDD include social anxiety biases although these social concerns are unlikely to occur early in life during adolescence or young adulthood.
- Body dysmorphic disorder can also present with a high degree of social anxiety although this is keenly associated with shame due to the patient's beliefs about body shape and weight.
- Narcissism can also present with social anxiety due to a fear of not being recognised as superior; this is categorically different to a SAD patient's fear of social inappropriateness.
- Avoidant personality disorder (AvPD) and SAD are often disputed as the same disorder but in AvPD it can be partially differentiated by the patient's belief that judgement is justified.
- Many eating disorders are also associated with social anxiety possibly due to body image and weight control.

INITIAL ASSESSMENT AND MANAGEMENT

During the initial assessment, clinicians should establish a good relationship so as to determine whether the primary diagnosis is SAD. Co-morbid disorders that may affect short- and long-term treatment goals should be identified during assessment. Furthermore, the clinician should prepare the groundwork for an appropriate treatment strategy.

Important points of a comprehensive assessment should include the following points [Andrews et al 2018]:

- Assess the severity and duration of symptoms and has the patient received prior treatment?
- Level of distress or functional impairment is the patient experiencing?
- Does the patient have any co-morbid depressive or anxiety disorders, substance abuse disorders, or medical health issues?
- Mental health history of the patient and family?
- Quality of the patient's interpersonal relationships and support network and are these affected by employment status or living conditions?
- The goals the patient expects from treatment.

CLINICAL RECOMMENDATIONS

Improved lifestyle factors such as regular exercise, healthy eating habits, and improved sleeping routines should be encouraged.

The reduction in caffeine, tobacco, and alcohol intake can help promote these new lifestyle behaviours [Andrews et al 2018].

TREATMENT

A watchful waiting approach to therapy is advised to monitor the efficacy of psychoeducation as well as the patient's uptake of recommended lifestyle improvements [Andrews et al 2018]. If treatment is indicated, clinical practice guidelines emphasise that psychotherapy is the first-line choice when congruent with patient preferences:

- Administration of CBT by a clinician or a program of guided digital CBT (dCBT) is advised for those patients with mild SAD.
- Treatment of moderate or severe SAD can be approached using a combination of CBT and pharmacotherapy with a SSRI or SNRI.
- Research shows that paroxetine, sertraline, fluoxetine, escitalopram, and fluvoxamine as well as the SNRI, venlafaxine, have been previously used in the treatment of SAD [Ipser et al 2008; Canton et al 2012].
- After 4 to 6 weeks of medication (or 4 to 6 sessions of weekly CBT), treatment response should be evaluated: continue therapy if there is a full or partial response; modify or escalate treatment if there is an inadequate response.
- The treatment strategy for those patients that aren't responding should consider the addition of a SSRI antidepressant if not already being administered, changing dCBT for CBT with an experienced clinician, increase the antidepressant dose if current dose is tolerated.

MONITORING PROGRESS

A second review at week 8 to 10 can be performed to consider whether there is continuing distress and whether treatment should be further modified: increase CBT frequency and/or combine CBT with a SSRI or SNRI. Furthermore, clinicians are encouraged to check for diagnostic accuracy, level of SAD severity, the presence of comorbidities, and to seek a second opinion [Andrews et al 2018].

- A delayed response to treatment is possible as it can take up to 12 weeks before an adequate response is observed. However, this assumes that there has been good adherence to the full therapeutic dose and that the patient can tolerate any ongoing side-effects if treatment is to be continued.
- Antidepressant medication can be changed to a different SSRI or it can be switched to a SNRI antidepressant. However, if both a SSRI and a SNRI have already been sufficiently trialled then a MAOI such as phenelzine can be administered after a washout period [Davis et al 2014].
- Neuropathic pain drugs, pregabalin and gabapentin, can be considered at this stage and although these have been rationalised as off-label anxiolytics for anxiety-related disorders there is a lack of evidence for efficacy in these psychiatric indications.
- Finally, benzodiazepines can also be trialled however it is advised that these are not to be considered as a long-term option in SAD. Beta-blockers, buspirone, and anti-psychotics should be avoided for the treatment of SAD.

SUMMARY

The persistent fear of social situations is a pervasive and distressing disorder that can be situation specific or more generalised. SAD is often in the presence of other comorbid disorders, which complicate diagnosis and treatment. Treatments that have an educational foundation can help address the psychological basis of SAD although psychotherapy with CBT is encouraged. Combination therapies with pharmacological interventions may be indicated due to individual comorbid disorders.

OBSESSIVE COMPULSIVE DISORDER

The hallmark of obsessive-compulsive disorder (OCD) is the presence of obsessions and compulsions. It has a bimodal incidence with peaks occurring in late childhood/early adolescence and again in early adulthood (20-29).

Obsessions

1. Persistent, unwanted, repetitive, intrusive thoughts, images, and urges which are ego-dystonic and cause anxiety and distress

Compulsions

- Repetitive behaviours or mental acts performed in response to an obsession to reduce the anxiety or distress to prevent a feared consequence.

TYPES OF OBSESSIONS AND COMPULSIONS

Obsessions	Compulsions
Fear of contamination	Cleaning or washing rituals
Pathological doubt	Repetitive checking
Sexual or Violent intrusive thoughts	Repetitive undoing thoughts
Fear of causing harm	Repeated checking
Need for symmetry and exactness	Ordering or arranging things
Religious obsessions	Religious rituals e.g. excessive praying
Superstitious obsessions	Superstitious rituals e.g. repeating activities a certain number of times

Screening

Recently validated screening tools include the Yale-Brown Obsessive-Compulsive Scale (Y-BOCS), which has good reliability and is widely used in clinical research.

Treatment

Although a systematic review and meta-analysis showed that cognitive behavioural therapy (CBT) has been shown to be more effective than selective serotonin reuptake inhibitors (SSRIs), in clinical practice both SSRIs and CBT are often used together.

Both individual and group CBT are as effective as each other in the treatment of OCD with the most common psychotherapeutic approach being exposure-response prevention CBT (ERP-CBT).

Exposure-response prevention CBT is a structured form of psychotherapy whereby the patient is exposed to situations that would provoke the obsessions and associated anxiety and distress and then instruct the patient to resist the associated compulsions or avoidance behaviours.

The psychological mechanisms underpinning this technique are habituation and extinction.

- Habituation is the decline in response after repeated exposure to a stimulus.
- Extinction refers to the cessation or reduction of behaviour when a reinforcer is taken away or ceases.

Exposure-response prevention CBT is advised to have a frequency and duration of 13 to 20 weekly sessions or 3 weeks of daily sessions. If necessary, additional sessions can then be offered to the patient 3 to 6 months after the first session.

PHARMACOTHERAPY

SSRIs

SSRIs are considered the first line in the treatment of OCD.

Trials have shown no difference between any particular SSRI in treating OCD symptoms.

High dose SSRIs have been shown to be more effective than lower doses. The medication and target doses are

- Fluoxetine - 80 mg/day
- Fluvoxamine - 300 mg/day
- Sertraline - 200 mg/day
- Paroxetine - 60 mg/day
- Citalopram - 40 mg/day
- Escitalopram - 40 mg/day

SSRIs should be prescribed for at least 12 weeks to determine responsiveness.

Approximately 25% of patients with refractory OCD to SSRIs may benefit from up to 28 weeks treatment after initiation.

SSRIs should be prescribed for at least 6-12 months after response to prevent a relapse.

Clomipramine

The tricyclic antidepressant, clomipramine, has larger effect sizes for a response than SSRIs.

It can be started at 25 milligrams per day. A target dose of 250 milligrams per day is recommended.

However, patient tolerability is poor due to significant anticholinergic side effects (dry mouth, blurred vision, fatigue, tremor and hyperhidrosis).

Therefore, based on safety data, clomipramine should instead be used as a second-line agent.

TREATMENT REFRACTORY OCD

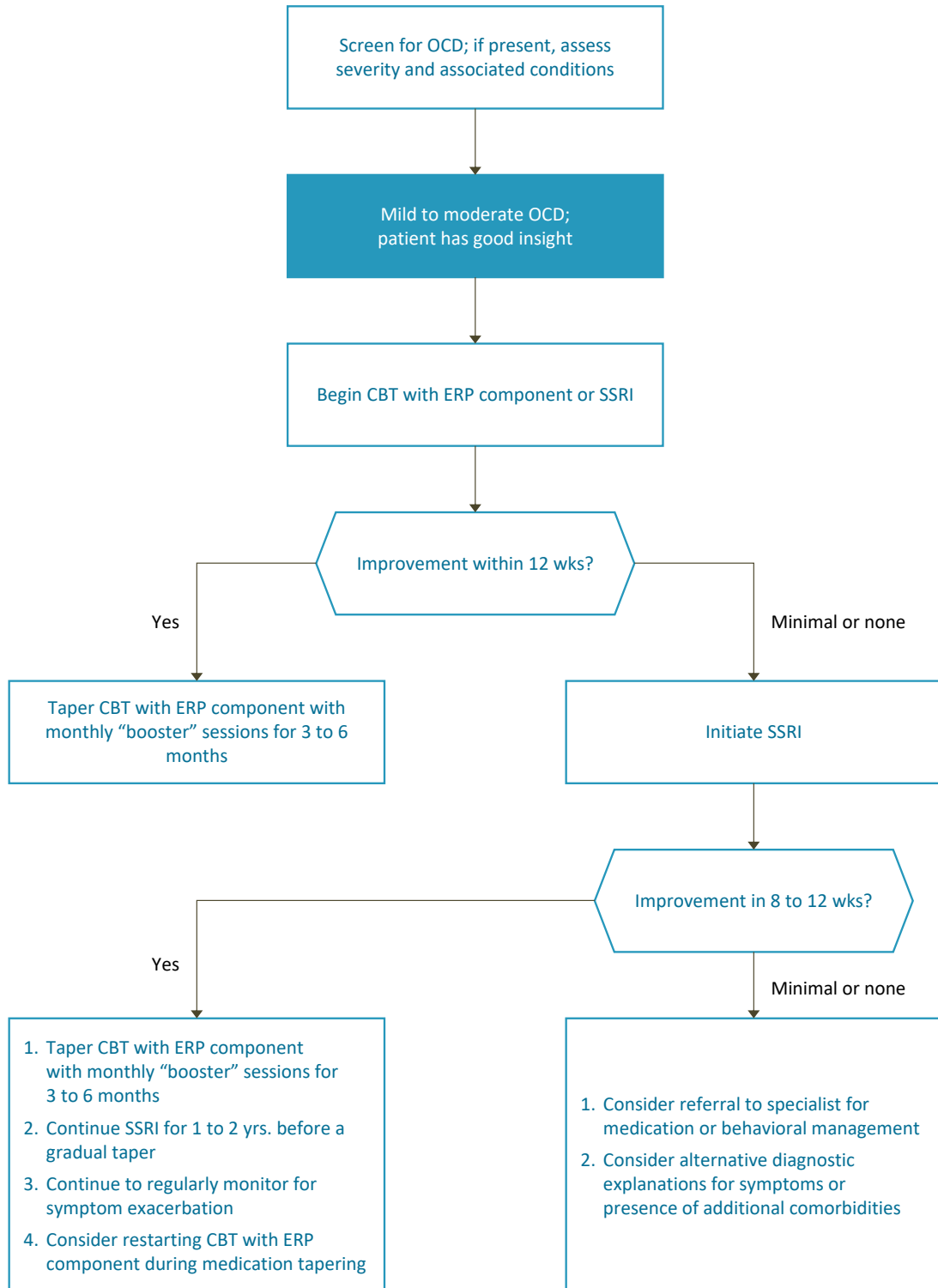
Approximately 25% of all OCD patients will be classified as treatment-resistant: this is defined as failing two different SSRI courses.

Furthermore, 40-60% of all patients may be considered as responding positively to treatment but will have ongoing residual symptoms.

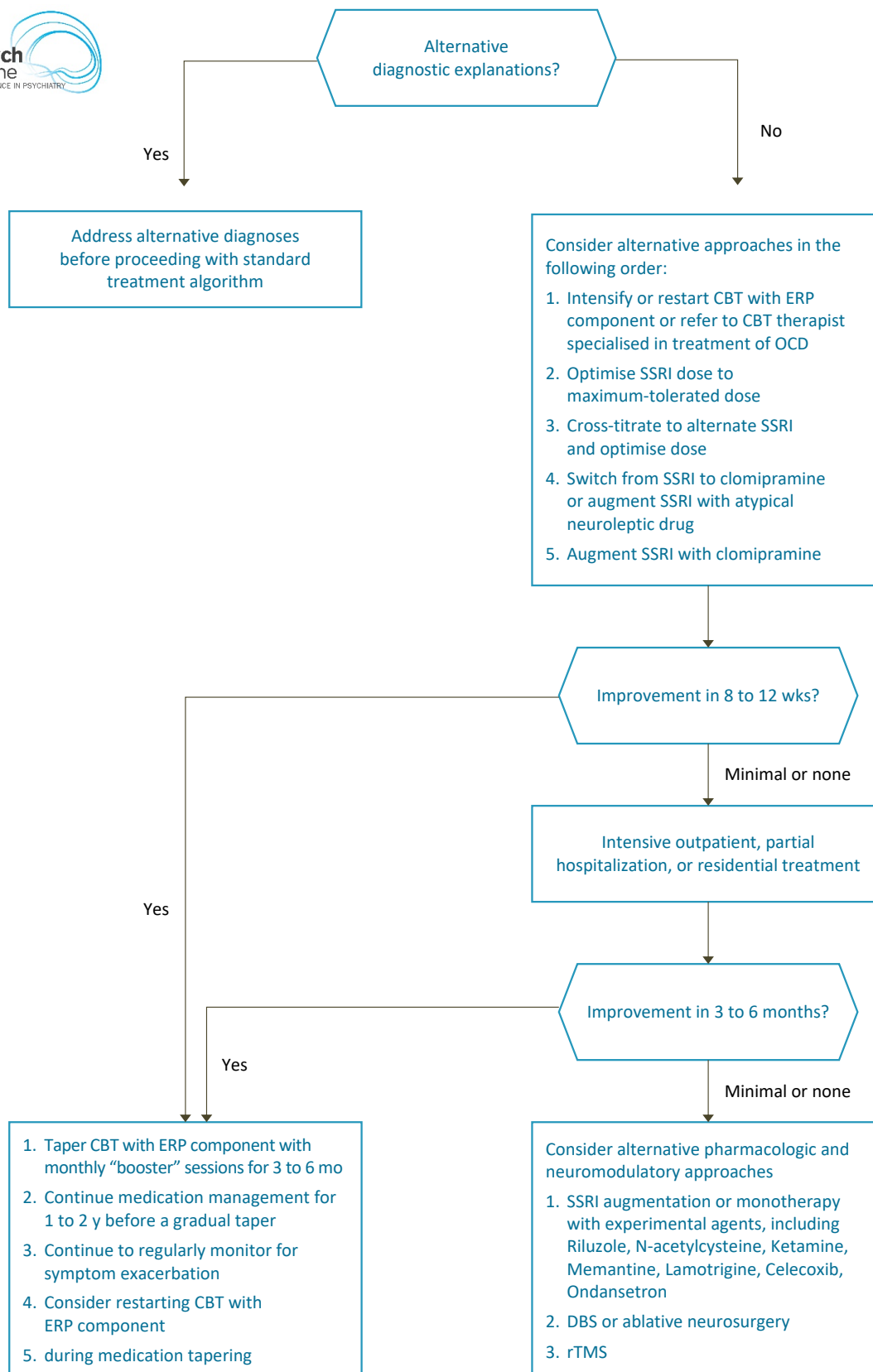
Augmentation with antipsychotics is effective in 30% of treatment-resistant patients.
Haloperidol is an effective augmentation agent in OCD with comorbid tics.

- Aripiprazole 5-10 mg/day
- Risperidone 1-3 mg/day
- Haloperidol 2.5-10 mg/day
- Olanzapine 5-10 mg/day

Treatment Algorithm for Obsessive Compulsive Disorder in Adults



Hirschtritt, M. E., Bloch, M. H., & Mathews, C. A. (2017). Obsessive-compulsive disorder: advances in diagnosis and treatment. *Jama*, 317(13), 1358-1367.



Hirschtritt, M. E., Bloch, M. H., & Mathews, C. A. (2017). Obsessive-compulsive disorder: advances in diagnosis and treatment. *Jama*, 317(13), 1358-1367.

SCHIZOPHRENIA

RISK

1. Suicide Risk – ranges from 20% to 50%

Risk factors for suicide in schizophrenia (Pompili 2007)

- Young
- Single
- Caucasian
- Depression and hopelessness
- Previous suicide attempt
- Social isolation
- Drug and alcohol misuse
- Insight retained
- Good pre-morbid function
- Deteriorating physical function
- Within first week after discharge
- Akathisia
- Unemployment
- Hospitalisation close to roads, railway
- Negative attitudes towards medication

2. Homicide

- Male
- Previous offences
- Lack of thought
- Threat control override symptoms (persecutory delusions, passivity phenomenon and command hallucinations)
- Check HCR-20

3. Water intoxication

Most cases of water intoxication occur in schizophrenia. Clinical symptoms include blurred vision, vomiting, tremor, confusion and psychosis. More severe symptoms include delirium, coma, and convulsions.

The exact pathophysiology is unclear. It is understood to be linked to psychotic delusions – leads to SIADH

4. Non-compliance

Patients on antipsychotics

- 30% of patients comply poorly with medication regimes, especially in schizophrenia.
- <50% of patients are compliant with medication one month after discharge, and <30% at two years (Remington & Collins)
- Clinicians were wrong about 55% of time in predicting compliance.

- Compliance is not strongly related to side-effects of medication or lack of insight into illness but rather to issues such as therapeutic alliance and family and social support (Diaz).
- Both patients and clinicians overestimate adherence. (J. Psych Neuroscience 2001)

Stopping medication is the most powerful prediction of relapse.

COMMONEST REASONS FOR NON-COMPLIANCE

Patient factors	Clinician factors	Medication factors	Illness factors	System factors
Attitudes to medication Stigma Family attitudes Cultural beliefs	Therapeutic alliance	Complexity of regime Side effects Lack of efficacy	Hallucinations Delusions Depression Demoralisation	Cost of medication Travel required

Treating non-compliance

- Simple strategies – explanation, reassurance, Dosette box, alarm clocks, Post-It notes, etc.
- Depot medications
- Psychoeducation
- Cognitive Behavioural Therapy (CBT)
- Interpersonal Psychotherapy (IPT)
- Marital/family therapy

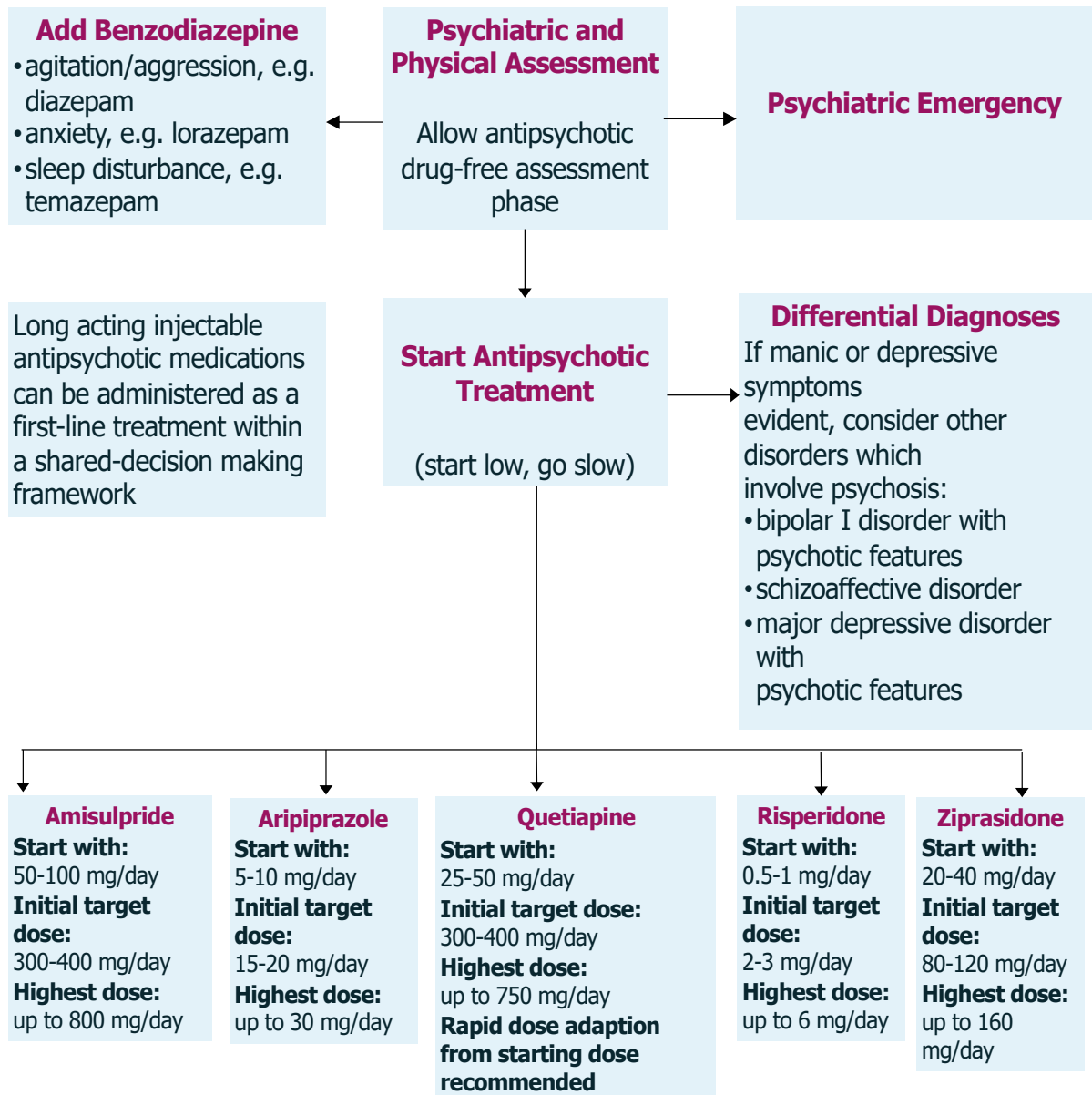
5. Risk of Substance Misuse

TREATMENT OF SCHIZOPHRENIA

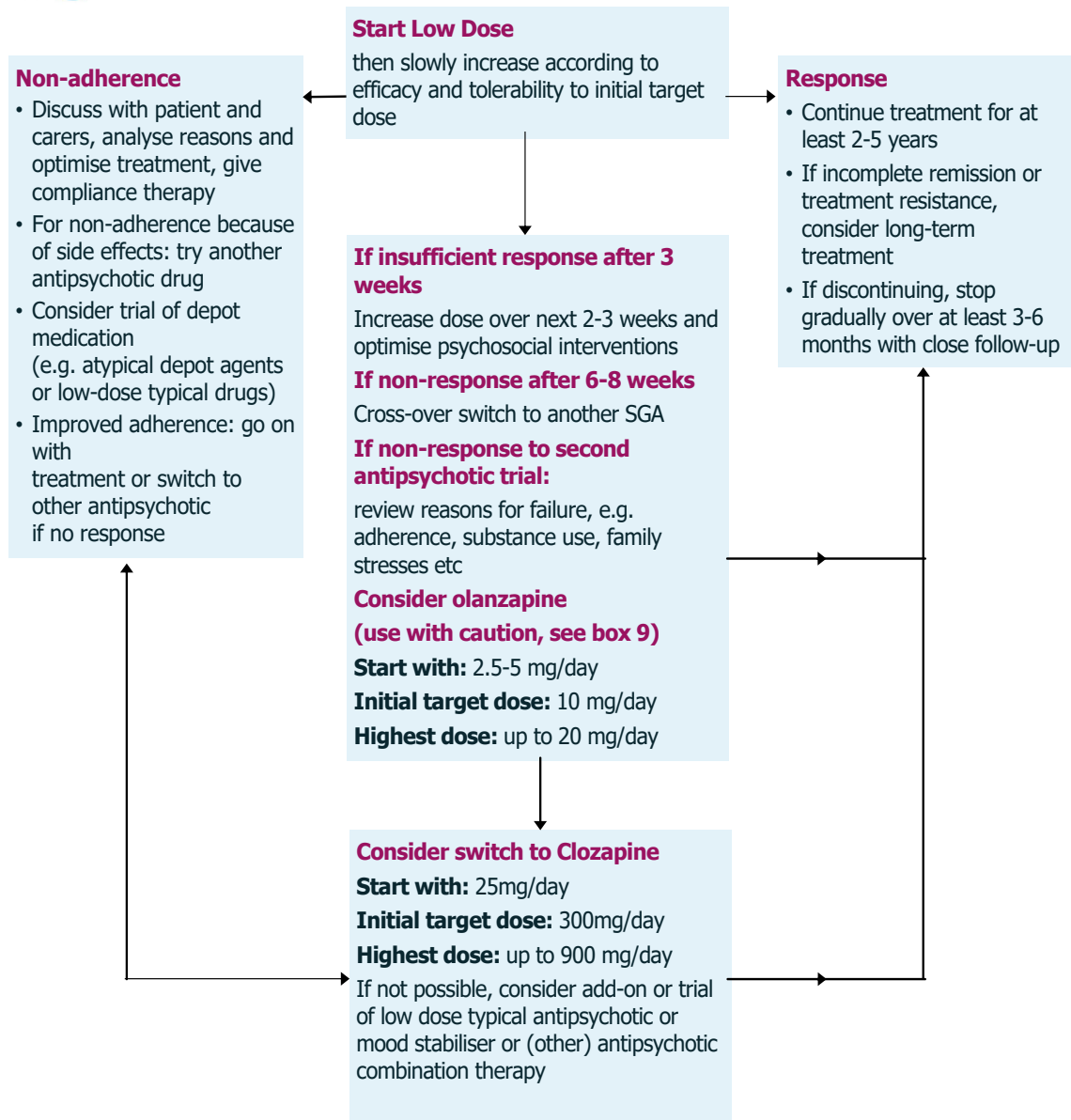
Pharmacological treatment

- The mainstay of treatment is typical and atypical antipsychotics.
- Information from CATIE Trial: Independent study sponsored by NIMH:
 - Phase I – Sample of 1493 patients randomly assigned to medication with one of five blinded antipsychotics for up to 18 months; Perphenazine (typical), Olanzapine, Quetiapine, Risperidone or Ziprasidone
 - Phase II – Sample consisted of discontinued patients before 18 months.
 - Sample of 543 patients randomly assigned to one of five second generation antipsychotics not received during Phase I (open label)
 - Results: (Relevant for exam): 74% patients discontinued before 18 months; Clozapine and Olanzapine had longest all cause discontinuation times i.e. stayed on these antipsychotics for the longest; the most frequent reason for discontinuation by patients taking Olanzapine was weight gain.

PHARMACOLOGICAL TREATMENT FOR FIRST-EPISEDE NON-AFFECTIVE PSYCHOSIS (RANZCP GUIDELINES)



Galletly, C., Castle, D., et al., 2016). Royal Australian and New Zealand College of Psychiatrists clinical practice guidelines for the management of schizophrenia and related disorders. *Australian & New Zealand Journal of Psychiatry*, 50(5), 410-472.



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PSYCHOSOCIAL INTERVENTIONS

1. Family interventions
2. Psychoeducation
3. Cognitive Behavioural Therapy
4. Social skills training
5. Vocational rehabilitation
6. Case management
7. Cognitive remediation

1. Family interventions

- Building alliance with the family
- Educate family about illness – High Emotional Expression
- Specialised family therapy –
 - enhancing communications
 - setting boundaries
 - promoting independence
- Evidence – effective in reducing relapse
 - Good evidence
 - Multi-family groups are better than single family groups

2. Psychoeducation

- For patient, family, carers and other community agencies, Number Needed to Treat – 9

3. Cognitive behavioural therapy

Cochrane Review showed evidence for the following:

- Negative symptoms
- Useful for resistant delusions and hallucinations

4. Social skills training

Efficacy: A series of meta-analyses concluded that skills can be learned and maintained in an artificial setting but generalisation to the community rarely occurs spontaneously, it must be programmed (Schiz Bull 2006)

5. Vocational rehabilitation

- Supported employment is better than pre-vocational training.
- Reduces re-hospitalisation and improves insight

6. Act & case management

The Cochrane review showed that Assertive Community Treatment (ACT) is better than Case Management (CM).

7. Cognitive remediation

Focuses on the specific cognitive deficits, e.g., poor memory, difficulty in planning and decision making. It consists of three main components:

- Assessment
- Engineering treatment to remediate impairments
- Cognitive mediation.

TREATMENT RESISTANT SCHIZOPHRENIA

Defined by at least two failed treatment trials with two different antipsychotics.

Recent RANZCP guidelines suggest treatment with two different atypical antipsychotics. (Guidelines prefer atypical antipsychotics to be trialled as opposed to typicals)

- Incidence – 25% (Meltzer 1992)
- Clozapine is the drug of choice (Kane 1994)

Correlates of Treatment Resistance

- Perinatal complications (Wilcox, Nasrallah 1987)
- Obstetric complications

One of the most common causes of resistant schizophrenia is pseudoresistance

Rule out Pseudoresistance

Causes of Pseudoresistance

1. Extra Pyramidal Side Effects (EPSE)
2. Non-compliance with medications (discussed above)
3. Drug misuse
4. Pharmacokinetic effects – Fluvoxamine, smoking:
 - The prevalence of smoking in patients with schizophrenia is 71% in males and 44% in females. Thus, tobacco smokers have a lower serum concentration of haloperidol, olanzapine and clozapine, especially clozapine, as it is metabolised by CYP1A2. Caffeine also affects levels by inducing the same enzyme.
 - Rifampicin, carbamazepine and phenytoin – CYP3A4 metabolises quetiapine, aripiprazole and risperidone and to a lesser extent ziprasidone
 - Also relevant is a predisposition for rapid metabolism occurring in those with CYP2D6 and most South Europeans, Saudi Arabians and Ethiopians have this.

Managing pseudoresistance

- Optimize dose of antipsychotic medication
- Serum levels
- Check for pharmacokinetic and pharmacodynamic interactions
- Check EPSEs
- Screen for comorbidity, anxiety, depression, OCD and/or substance abuse
- Review organicity
- Assess compliance

MANAGEMENT OF TREATMENT RESISTANCE

1. Clozapine

Clozapine has been proven to be the most effective treatment for treatment-resistant schizophrenia (Kane 1992, Iqbal 2003)

In the CATIE trial, patients who received clozapine after discontinuation of two previous antipsychotics were less likely to discontinue treatment due to inadequate therapeutic response than patients who received quetiapine, risperidone or olanzapine

Side effects: Weight gain, Sedation, Sialorrhea, Convulsions, Myocarditis

The most important reason is inadequate duration of treatment. Meltzer concluded that 30% of patients would respond by 6 weeks, a further 20% by 3 months and 10–20% by six months. Therefore, monotherapy should be tried for 6–12 months.

- Check levels $\rightarrow$ 350 ng/ml
- Adjunctive drugs that may affect pharmacokinetics

Clozapine is augmented by:

- Amisulpride (Zink 2004)
- Sulpiride (Shiloh 1997)
- Antiepileptics, Lamotrigine or Topiramate (Tiihonen 2003)
- Sodium Valproate (Basan 2004; Cochrane)
- Lithium (caution increases neurotoxicity (Small 2003)
- Carbamazepine (Caution) (Leicht 2002)
- Fluoxetine – pharmacokinetic effects by enzyme inhibition
- ECT (Israeli study but evidence is not conclusive)
- Some evidence for Omega 3 and Omega 6 (Cochrane library)
- Omega 3, short-term benefit but not long term
- Cognitive deficits – Donepezil
- RTMS for auditory hallucination (Fitzgerald 2005, Hoffman 2005)

MANAGEMENT OF NEGATIVE SYMPTOMS

15-25% of people with schizophrenia report negative symptoms

Primary negative symptoms

- Affective flattening
- Poverty of speech (Alogia)
- Diminished drive (Amotivation)
- Loss of pleasure (Anhedonia)

Secondary negative symptoms

- Positive symptoms
- Depression or demoralisation
- Post-schizophrenia exhaustive syndromes
- Substance misuse behaviour
- EPSE

TREATMENT OF PRIMARY NEGATIVE SYMPTOMS

Pharmacological – newer antipsychotics

- Glycine – mixed results
- D-Cycloserine
- Fluvoxamine
- Mirtazapine
- Galantamine
- Pergolide

Psychosocial therapies

Provide greater benefit on negative than positive symptoms

- Psychoeducation – group and family
- Social skills training (Kopelowicz 1997)

- Cognitive Behavioural Therapy (Sensky, Turkington 2000)
- Cognitive remediation (Wykes 1999)
- Group family therapy (Dyck)

MANAGEMENT OF EXTRA PYRAMIDAL SIDE EFFECTS (EPSE)

DYSTONIA

- Typical antipsychotics, metoclopramide can cause dystonic reactions
- Benztropine (anticholinergic) is treatment of choice (1-2 mg I.M.)

PSEUDOPARKINSONISM

- Pseudoparkinsonism is characterised by tremor, bradykinesia, rigidity, bradyphrenia
- Differentiated from Parkinson's disease by presence of bilateral tremor and rigidity
- Pseudoparkinsonism can be assessed by the Simpson-Angus rating scale
- Risk factors for Pseudoparkinsonism include being an elderly female, and pre-existing neurological impairment
- Management includes reduction of dose, change to atypical if on typical and prescribe anticholinergic (e.g. Benztropine)

AKATHISIA

- Measured by Barnes Akathisia Scale
- Increases suicide risk
- Management in following order:
 - Reduce dose
 - Change to atypical if on typical
 - Benztropine may help if other Parkinsonian symptoms present
 - Propranolol
 - Cyprohepatadine
 - Benzodiazepine
 - Clonidine

TARDIVE DYSKINESIA

- Lip smacking, tongue protrusion, choreiform hand movements, pelvic thrusting
- Measured by AIMS
- Risk factors include diabetes, typical antipsychotics, organic neurological impairment, female, diabetes, smoking
- Management in following order
 - Stop anticholinergic
 - Reduce dose of antipsychotic
 - Change to atypical drug if on typical
 - If untreated, Clozapine

Other strategies include:

Tetrabenazine – depressogenic, useful in Huntington's
 Benzodiazepine
 Vitamin E

Neuroleptic Malignant Syndrome (NMS)

- Fever, diaphoresis, rigidity, fluctuating consciousness (delirium), autonomic instability (fluctuating BP, tachycardia)
- Investigations: Increased CK, leukocytosis, altered liver function tests
- Risk factors: High potency typical depot
Abrupt dose increase or decrease
Abrupt withdrawal of anticholinergics
Parkinson's disease, organic brain disease
Hyperthyroidism, dehydration, psychomotor agitation, intellectual disability

Management in the following order:

- Medical team to be called as medical emergency
- Withdraw antipsychotics, monitor Temperature, Pulse, Blood Pressure
- In medical unit: rehydration, Bromocriptine and Dantrolene, benzodiazepines, cooling blankets, artificial ventilation if required
- ECT
- Antipsychotics may be restarted with caution
- Start low and go slow with monitoring of CK, T, P, BP
- Consider antipsychotic with low dopamine affinity, e.g. Quetiapine, Olanzapine

Side Effects of Clozapine

Haematological side effects of clozapine

- Agranulocytosis: Risk factors include Afro-Caribbean, youth. Not dose related
- Neutropenia
- Thromboembolism and pulmonary embolism

Cardiovascular side effects of clozapine

- Myocarditis – Flu-like symptoms, tachycardia at rest, fever, fatigue, dyspnoea and chest pain. ECG changes, eosinophilia and enlarged heart on CXR

Management of myocarditis:

ECG, Echocardiogram

Refer to cardiologist. If signs of heart failure stop clozapine immediately

- Cardiomyopathy
- Tachycardia
- Prolonged QT
- Pericarditis
- Pericardial effusion

- Cardiac failure
- Mitral valve insufficiency

Metabolic syndrome

- Central obesity
- Raised serum triglyceride level: ≥ 150 mg/dl
- Reduced serum HDL cholesterol: ≤ 40 mg/dl in males and ≤ 50 mg/dl in females
- Raised BP: Systolic BP ≥ 130 mm/hg or diastolic ≥ 85 mm/hg
- Raised fasting plasma glucose: ≥ 100 mg/dl or previously diagnosed type-2 diabetes

Management of metabolic syndrome

Baseline monitoring:

- Personal and family history of obesity, DM, dyslipidaemia, hypertension or CVD
- Weight and height (calculate BMI)
- Waist circumference at level of umbilicus
- Blood pressure, fasting blood glucose, fasting lipids (LDL, HDL, Cholesterol and Triglycerides)
- Treatment with the Second-Generation Antipsychotics has least weight gain potential
- Educate family and patient about side effects and signs and symptoms of diabetes
- Educate patient about importance of exercise and cessation of smoking and other modifiable risk factors

Follow up and monitoring

- Monitor BMI and waist: hip ratio every visit or 3–6 monthly
- Monitor blood glucose, lipids and BP 3–6 monthly

Treatment of weight gain and metabolic syndrome (Rege 2008)

- Refer dietician
- Exercise regime and diet modification
- Treat diabetes and hyperlipidaemia accordingly in conjunction with endocrinologist
- Switch antipsychotics to lower weight gain potential, e.g. Aripiprazole, Ziprasidone
- Other biological strategies include Orlistat (lipase inhibitor), Sibutramine (monoamine re-uptake inhibitor), Rimonabant (cannabinoid receptor antagonist), Topiramate and Metformin

FIRST-EPISODE SCHIZOPHRENIA

Prodromal states (Yung & McGarry 1996)

1. **Attenuated:** sub-threshold psychotic symptoms over one week to five years
2. **Trait schizotypal disorder or first relative with schizophrenia** plus decrease in Global Assessment of Functioning over one week to five years.

3. **BLIPS** (Brief Limited Intermittent Psychotic Symptoms): At least one psychotic symptom for less than one week in past year.

Any one of the following moderately predicts psychosis within twelve months:

- Trait and attenuated
- Trait or attenuated less than five years
- Trait or attenuated or BLIPS + GAF less than 40
- Family history first and functional deterioration
- Schizotypal personality and recent functional deterioration
- 20–40% conversion.

Longer duration of untreated psychosis is associated with

- poor psychosocial function
- high suicide rate
- treatment resistance
- drug misuse

PRINCIPLES IN MANAGEMENT OF FIRST-EPISODE PSYCHOSIS

- Early detection and assessment of frank psychosis
- Specific focus on therapeutic engagement
- Comprehensive assessment
- Embracing diagnostic uncertainty
- Treatment in least restrictive setting using low dose medication

PHARMACOTHERAPY

- Antipsychotic free observation period
- Low threshold for use of Antipsychotics
- Short term Benzodiazepines + Atypical Antipsychotics
- Maintain treatment for one year

Cognitive Behavioural Therapy

Timing needs to be individually tailored (Chiles 1995)

Psychosocial and family interventions

Treating depression and anxiety – post-depression years

DEPRESSION

- Lifetime prevalence – around 20%

CO-MORBIDITY WITH DEPRESSION

- Co-morbidity is more the rule than the exception, e.g. in the National Comorbidity Survey (NCS) and ECA, 74% and 75% respectively of patients with major depression had other lifetime diagnosis.

Three classes of co-morbidity occur most commonly

- Substance misuse
- Anxiety disorders
- Personality which contributes to poor treatment response (Schulberg)

The WHO study of primary care sites (Ustun & Sartorius 1995) found that current co-morbidity stands at around 62%.

1. RISK

Suicide

- Epidemiological Catchment Area (ECA) study – lifetime rate of suicide attempts was 7.9%, with 19.5% in those with co-morbid alcohol abuse/dependence compared with 1% in those without documented psychiatric illness.
- Harris and Barraclough (1998) completed a series of meta-analyses that showed that risk of suicide was increased by about 21 times (similar to Bipolar illness)
- Major depression also accounts for 5.1% of the total global burden of disease with respect to Disability Adjusted Life Years (DALY's) lost to individual and society.

Risk factors for Depression

- Women are at greater risk by 1.5 – 3 times that of men (Robins & Regier)
- Age: two peaks (early and late onset)

Kindling: Personal history of mental disorder → previous episodes may increase the risk through sensitising the individual through the Kindling phenomenon → lower and lower levels of stress are required to provoke new depressive episodes and ultimately such episodes may occur spontaneously (Post 1992)

Genetics

- 30–54% in monozygotic twins
- 20% in dizygotic twins
- 5–15% in non-twin population

Marital status: Unmarried or divorced > married

Socio economic status: Inverse relationship

Urbanicity: greater rates in urban areas

Physical illness: psychological and biological components

Harris & Brown Stressors for depression in women:

- Loss of mother before age 11
- Three or more children under the age of 14
- Lack of confidant
- Unemployed

2. TREATMENT OF DEPRESSION

RANZCP Guidelines

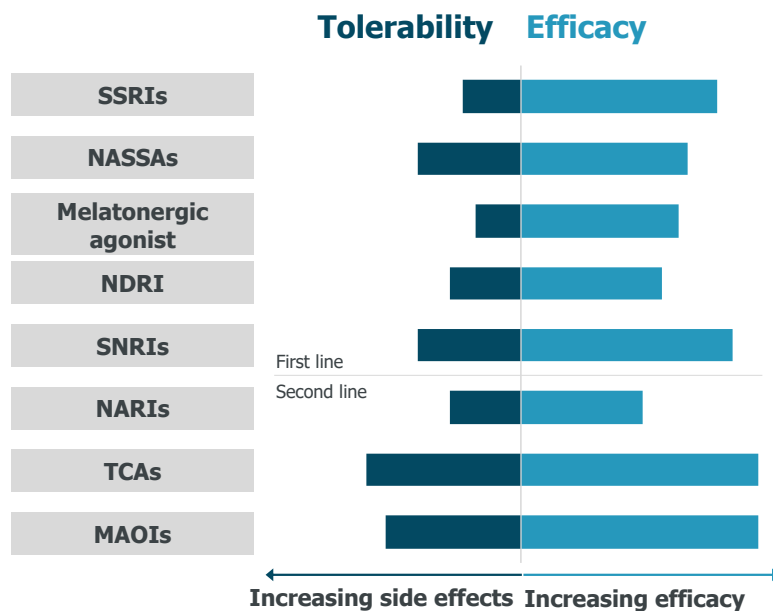
Mild depression

- Antidepressants are not recommended as first line treatment
- Watchful waiting
- Guided self-help – written materials and limited support
- Psychological interventions – brief CBT, problem solving and supportive counselling
- Structured and supervised exercise programme of 10–12 wks. duration, 3 sessions /week with each session lasting for 45 minutes to 1 hour
- Sleep and anxiety management
- Antidepressants are not advised for mild depression but may be used if past history of severe depression and if persists after all the above.

Moderate depression

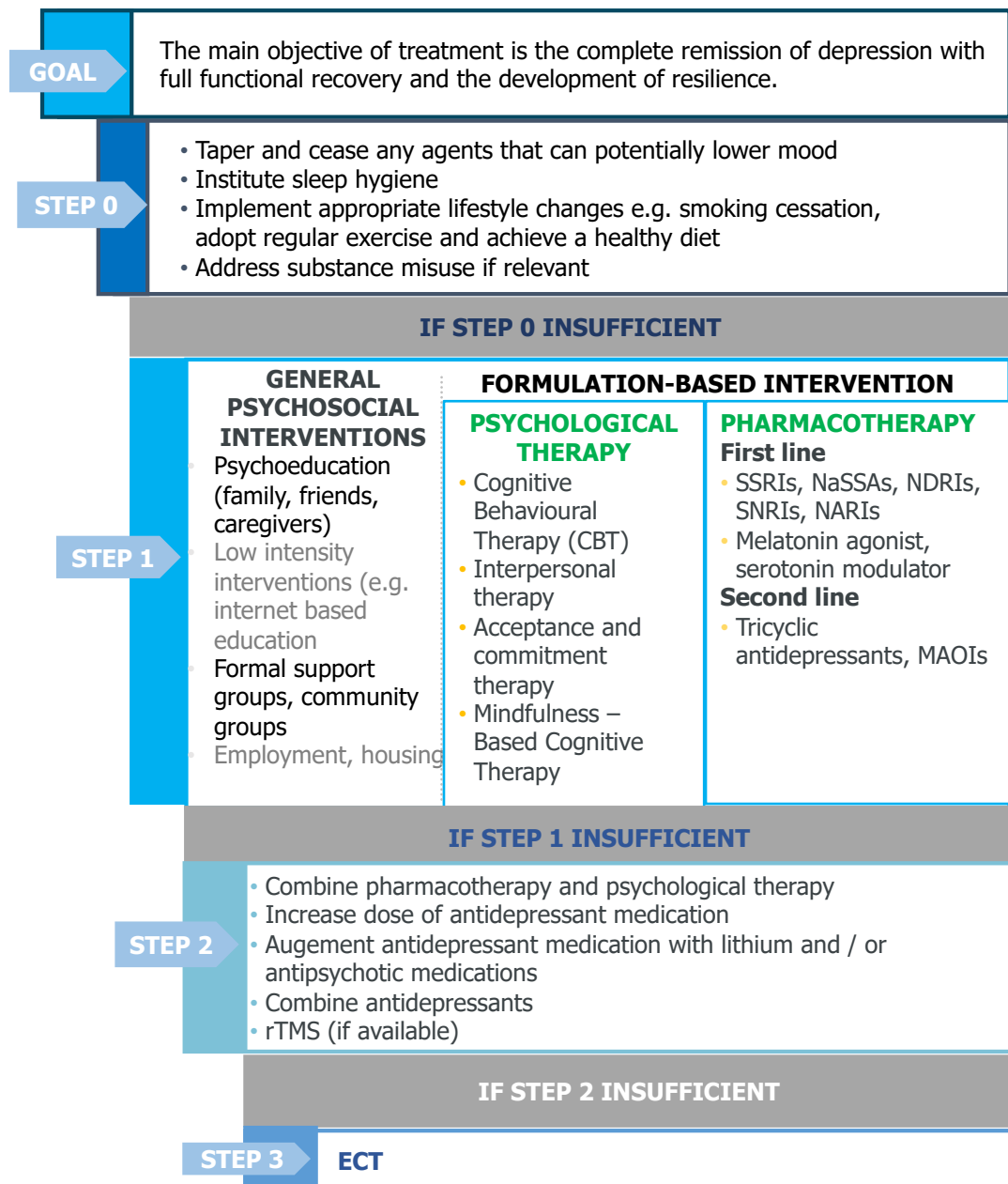
- First line is SSRI/TCA, Venlafaxine or Mirtazapine (not mentioned in RANZCP guidelines), depending on side-effect profile, pharmacokinetic and pharmacodynamic effects
- If no response, switch to another SSRI or any of the others. Trial of 6–8 wks. is necessary to confirm non-response.
- If no response then augment with lithium, level of 0.6–0.8 mmol/l
- If no response, then consider ECT after second opinion (see notes on management of treatment resistant depression, below)
- CBT is equally effective as medication in moderate depression but with slower onset of action and needs 6–9 sessions. Booster sessions are recommended after remission, in case of another relapse or for maintenance
- IPT is also recommended

CLINICAL UTILITY OF ANTIDEPRESSANTS



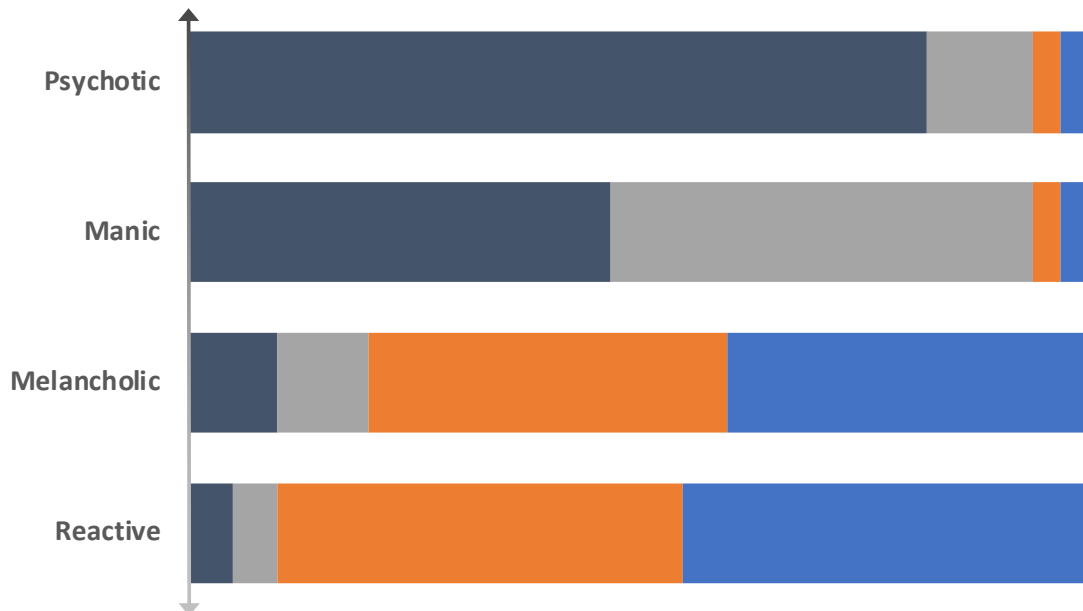
Malhi, G. S., Bassett, D., Boyce, P., Bryant, R., Fitzgerald, P. B., Fritz, K., ... & Porter, R. (2015). Royal Australian and New Zealand College of Psychiatrists practice guidelines for mood disorders. *Australian & New Zealand Journal of Psychiatry*, 49(12), 1087-1206.

MANAGEMENT OF MAJOR DEPRESSIVE DISORDER



Malhi, G. S., et al. (2015). Royal Australian and New Zealand College of Psychiatrists clinical practice guidelines for mood disorders. *Australian & New Zealand Journal of Psychiatry*, 49(12), 1087-1206.

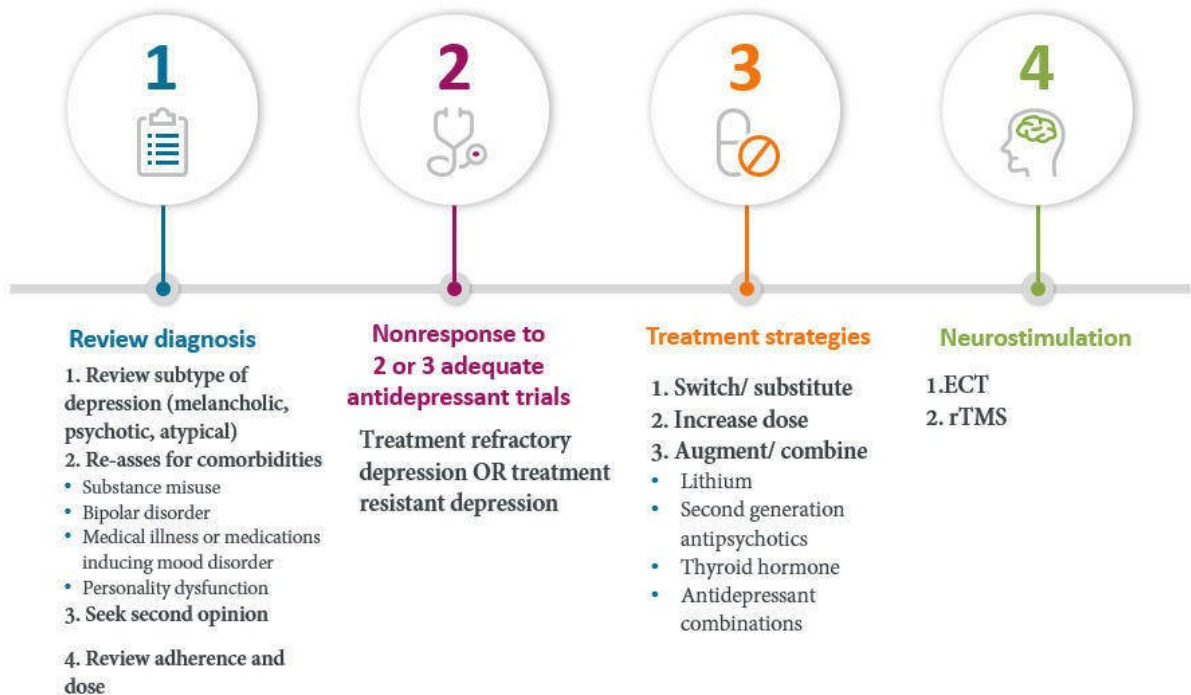
BIOPSYCHOSOCIAL AND LIFESTYLE MODEL



Biological Treatments	Psychological Treatments	Social Treatments	Lifestyle Treatments
• Antidepressants • Antipsychotics • Mood stabilisers • Electroconvulsive therapy • Transcranial magnetic stimulation	• Brief Cognitive behavioural therapy • Formal cognitive behavioural therapy • Interpersonal therapy • Mindfulness • Acceptance and commitment therapy • Schema therapy	• Family psychoeducation • Family/ friends • Formal Support groups • Community groups • Caregivers • Employment • Housing	• Exercise • Diet • Smoking cessation • Alcohol cessation • Managing substance misuse • Sleep

Malhi, G. S., et al. (2015). Royal Australian and New Zealand College of Psychiatrists clinical practice guidelines for mood disorders. *Australian & New Zealand Journal of Psychiatry*, 49(12), 1087-1206.

STRATEGIES TO ADDRESS MEDICATION NON-RESPONSE



TREATMENT RESISTANT DEPRESSION

There is no universal definition for treatment resistant refractory depression. At least two adequate monotherapy trials with drugs from different pharmacological classes fail to elicit therapeutic response is one definition. (APA.DSM 2003).

Prevalence is 10–30% (Joffe 1996).

Causes of Treatment Failure

- Inaccurate assessment, Bipolar, schizoaffective, ADHD, Axis II disorders
- Non-compliance
- Co-morbidity – substance misuse, Bipolar, personality disorders
- Inadequate dose, duration
- Psychosocial factors
- Medical condition undiagnosed → e.g. Hypothyroidism
- Drugs → B-Blockers, Methyl dopa

Bipolar: **Hints for bipolarity (WHIPLASHED)** (Pies, Curr Psych 2007)

- W** – Worse or wired when taking antidepressants
- H** – Hypomania, hyperthymic temperament or mood swings in history
- I** – Irritable, hostile or mixed features
- P** – Psychomotor retardation
- L** – Loaded family history: mood swings, bipolar affective illness
- A** – Abrupt onset and/or termination of depressive bouts < 3 months
- S** – Seasonal or post-partum depression
- H** – Hyperphagia and hypersomnia
- E** – Early age of onset
- D** – Delusions, hallucinations, other psychotic features

NOVEL TREATMENTS FOR DEPRESSION

Augmentation strategies: The use of a psychotropic agent to enhance the effect of an antidepressant through facilitating neurochemical effects.

- **Lithium** – odds ratio for improvement is 3.31 (Bauer 2000)
- **T3** – 4RCTs and odds ratio for improvement is 1.53 (Aronson 1996)
- **Thyroxine** (Lojko 2007)
- **Pindolol** (RANZCP guidelines)
- **Dopaminergic augmentations** – Pergolide, Amantadine, Pramipexole, Ropinirole
- No controlled studies but Pramipexole, Amantadine and Ropinirole have also been used to treat SSRI induced sexual dysfunction
- **Modafinil**
- **Atypical antipsychotic augmentations**
 - Risperidone
 - Olanzapine
 - Ziprasidone
 - Quetiapine
 - Aripiprazole
 - Olanzapine and Fluoxetine combinations (Thase 2007)
 - Aripiprazole +SSRI vs. Aripiprazole alone (Berman 2007: Robust 35% response 25% remission)
 - Quetiapine + SSRI
 - Risperidone
 - Methylfolate (15–30 mg/day) and S-Adenosyl –L-Methionine augmentation (800–1600 mg/day)
- **Anticonvulsant augmentations** – Limited
 - Lamotrigine (100–300 mg/day)
 - Gabapentin
 - Topiramate
 - Valproate
 - Omega 3 (1 g dose, not 2–4 g/day)
- **Riluzole** – NMDA receptor antagonist

- **Other** – Inositol – recent double-blind study failed (Nemets 1999)
 - Opiates
 - DHEA
 - Oestrogen

Cognitive Behavioural Therapy (CBT)

- Lower relapse rates than pharmacotherapy after discontinuation of acute phase treatment (54% in two years, 29% in one year)
- More effective than IPT in severe depression.

Interpersonal Psychotherapy (IPT) (Sullivan)

- Good evidence base for treatment of depression (Luty 2007)
- Particularly useful in depression associated with grief due to loss or illness, e.g. HIV, Diabetes,

VNS: Vagus Nerve Stimulation

Electroconvulsive Therapy (ECT)

- Indicated in psychotic depression (80% response rate) and severe suicidality

Important interactions between medications and ECT

- **Antidepressants** – some evidence of augmentation
- **TCAs** – seizures and caution advised in patients with concomitant cardiac illness
- **SSRIs** – Prolonged seizures
- **Benzodiazepines** – Increase seizure threshold. Stop before ECT.
- **Mood Stabilisers** – Consult neurologist if epileptic.
May need reduction of dose before treatment. Stop 24–48hrs before ECT
- **Lithium** – increases risk of post ECT delirium. For certain patients on lithium, careful risk-benefit analysis is required as ECT induced mania is more significant than delirium.

REPETITIVE TRANSCRANIAL MAGNETIC STIMULATION (RTMS)

Uses a powerful hand-held magnet to create a time-ranging magnetic field where a localised pulse magnetic field over the surface of the head depolarises underlying superficial neurons (George & Belmaker 2000)

This TMS produces powerful brief magnetic fields that produce electric currents in the brain. If these are delivered repetitively and rhythmically it is called RTMS.

In all meta-analyses of RTMS vs. sham, TMS was significantly (statistically) superior to sham TMS (prefrontal) 0.67 effect size (moderate–large) (29% vs. 7%) (Burt et al. 2002).

Large variations in delivery were reported.

Response rates vary from 25 to 45%. No cognitive side effects were reported.

Side effects:

- Muscle tension type headache
- Discomfort
- Scalp tenderness

RTMS vs. ECT – end treatment similar results (Janicak 2002)

Magnetic Seizure Therapy (MST) – still experimental

Deep Brain Stimulation (DBS):

Involves creating a small hole in the skin and passing a small wire into selected brain regions.

In 2002, the FDA approved implanting Deep Brain Stimulators in subthalamic nucleus and globus pallidus for treatment of Parkinsonism. Study in six patients with refractory depression postulated to modulate BA25 to reduce metabolic activity – sustained remission in 4 out of 6 (Mayberg 2006).

Stimulation of the sub-genial anterior cingulate white matter can effectively reverse symptoms in treatment resistant depression. There are also recent studies citing benefit of the procedure in severe OCD.

BIPOLAR DISORDER

FORMULATION MODELS

- Biological models
- Cognitive model
- Interpersonal social rhythm model
- Self-medication hypothesis
- Psychodynamic models

1. Risk

- Risk of self-harm/ suicide
- Risk of harm to others
- Financial risk
- Risk to reputation
- Corporate risk
- Risk of accidental self-harm- driving
- Risk of sexual exploitation, sexual indiscretion

2. Clarification of diagnosis

- Important to check levels of mood stabilisers – Lithium / Valproate
- Alcohol misuse –Gamma Glutamyl Transferase (GGT)
- Thyroid – lithium induced thyroid dysfunction/ thyroid dysfunction causing relapse
- Lipids, glucose as antipsychotics may be required
- Syphilis screen if suspected
- Rule out schizoaffective disorder through collateral/ notes

3. Treatment of acute psychological symptoms: Based on TIMA (Suppes 2005) and CANMAT guidelines (McIntyre 2007)

Acute Mania

- Valproate 20–30 mg/kg or lithium 400 mg bd titrated upwards to plasma level of 0.8–1.2 mmol/L or Olanzapine or Risperidone
Benzodiazepines may be used as adjuncts, e.g. Clonazepam 0.5 mg B.D
- **No response:** combine lithium, atypical antipsychotics or Valproate.
- **No response:** three drug combinations/Clozapine or above with Carbamazepine or Oxcarbazepine/ECT

Acute depressive symptoms

- If on lithium or valproate, optimise levels/continue antidepressant if on it prior to episode
- **No response:** Lamotrigine augmentation with lithium or Valproate (Caution drug interaction with Valproate – reduced dose of Lamotrigine required)
- Or Quetiapine 300–600 mg (BOLDER I and BOLDER II studies)
- Or Olanzapine-Fluoxetine combination
- **No response:** Add antidepressant / ECT if severe

- **No response:** MAOI, TCA, Atypical APs not previously included, pramipexole, inositol, stimulants and thyroid supplementation. Combinations of drugs from previous stages.

Maintenance Treatment – Hypomania/Mania

- Level 1 Patients with frequent, recent or severe mania- lithium
- Patients without severe, frequent or recent mania- Lithium, Valproate or Lamotrigine (LTG)
- Level 2 Alternative – Olanzapine, Aripiprazole
- Level 3 CBZ or Clozapine
- Level 4 Quetiapine, Risperidone, Ziprasidone
- Level 5 Typical antipsychotics, OXC, ECT

Maintenance treatment + Most recent episode of depression

- Level 1 Patients with recent and/or severe history of mania: Lamotrigine combined with antimanic agent
- All other patients: Lamotrigine monotherapy
- Level 2 Lithium
- Level 3 Combination of antimanic and antidepressant in past, including OFC
- Level 4 Valproate, CBZ, Aripiprazole, Clozapine, Olanzapine, Quetiapine, Risperidone, Ziprasidone
- Level 5 Typical antipsychotics, Oxcarbazepine, ECT

Points to remember

- Lithium has more antimanic properties
- Lamotrigine has greater effect for depression
- Valproate is more effective for mixed episodes

Psychological therapies

- CBT
- Interpersonal Social Rhythm Therapy (IPSRT)
- Family focussed psychoeducation

Social Interventions

- Accommodation
- Financial
- Relationships
- Employment
- Legal interventions: Driving, Restraining Orders, guardianship during illness, power of attorney

RAPID CYCLING DISORDER

- four or more episodes of mood disorder fulfilling criteria for mania or depression over a 12-month period
- It is also used to describe mood disorder that rapidly changes from one mood state to other without periods of remission

Risk factors for rapid cycling

- Subclinical hypothyroidism
- Drug and alcohol misuse
- Antidepressant medication
- Non-compliance with medication
- Female
- Bipolar II
- Organicity (Multiple sclerosis, Mental retardation)

Principles of management of Rapid cycling disorder

- Exclude non-compliance
- Treat any co-morbid substance misuse
- Exclude antidepressant induced affective instability
- Exclude subclinical hypothyroidism
- Trial alternative mood stabiliser alone or in combination
- Strongest evidence is for Lithium and Valproate combination

Main considerations in initiating lithium

- Narrow therapeutic index
- Renally excreted
- Not hepatically metabolised
- Start at dose of 250 mg BD or 400 mg nocte depending on formulation and measure levels after seven days and after one week after each dose change
- Levels measured 12 hrs post dose
- Once stable check three monthly
- Check renal and thyroid functions six monthly
- Levels aimed are between 0.6 and 1.2 mmol/l
- Sudden stopping of lithium increases the chance of rebound
- It is recommended that lithium treatment should not be started unless there is a clear intention that it is to be continued for at least three years

Main adverse effects of lithium and management

- Side effects are dose related and increase in frequency when levels are above 1 mmol/l
- Mild side effects - fine hand tremor, thirst and polyuria, nausea (transient)
- Acne and psoriasis can be worsened

Toxicity

- Levels >1.5 mmol/l
- GI effects – Anorexia, vomiting and diarrhoea
- CNS effects – muscle weakness, drowsiness, ataxia, coarse tremor, and muscle twitching

- Levels >2.0 mmol/l (emergency)
- Management in ICU with cardiac, renal monitoring
- Delirium
- Myoclonus
- Coma
- Osmotic diuresis or forced alkaline diuresis
- Above 3.0 mmol/l – peritoneal or hemodialysis

Endocrine side effects of lithium

- Hypothyroidism (more common in women)
- Hyperthyroidism
- Hypoparathyroidism
- Hyperparathyroidism

Renal complications of lithium (in order of frequency of occurrence)

- Small reduction in glomerular filtration rate
- Interstitial nephritis
- Reduction in urinary concentrating capacity (nephrogenic diabetes insipidus), reversible in short term but irreversible after long-term treatment

Important drug interactions of lithium

- Diuretics: thiazides most commonly associated with increased serum lithium levels as the kidney cannot distinguish between sodium and lithium at the proximal tubule
- NSAIDs increase lithium levels by about 40%
- ACEI and Angiotensin 2 antagonists increase the lithium level by decreasing excretion

Valproate

- Mood stabiliser
- Start at 20–30 mg/kg in acute mania, titrating to levels of 350–690 $\mu\text{mol/L}$ (50–100 mg/L), or depending on pharmacy
- Licensed for maintenance treatment of bipolar disorder

Lamotrigine

- Available but not on PBS for treatment of bipolar disorder in Australia but licensed by US FDA for maintenance therapy of Bipolar I.
- Dose dependent rash – Steven-Johnsons syndrome

Antipsychotics

- Several atypical antipsychotics are used for acute treatment of mania and olanzapine is licensed for prophylaxis.
- Quetiapine recently licensed for bipolar depression acute treatment as monotherapy based on BOLDER I and BOLDER II studies
- Aripiprazole is licensed for all phases by FDA

ORGANIC PSYCHIATRY

Important issues in formulation and management

- Risk of physical decompensation important
- Risk of suicide is important as physical illness is a significant risk factor
- Liaise with medical specialities
- Pharmacokinetic and pharmacodynamic interactions prominent
- Psychosocial adjustment due to illness
- Direct biological effects of illness e.g. neurological deficits in lobar domains presenting with particular symptoms
- Direct biological effects of medications causing psychiatric disorders
- Financial issues –cost of disability
- Occupational therapist involvement
- Legal issues of will and guardianship if capacity is an issue
- If terminal illness or progressively deteriorating illness then consider issues of advance directives, guardianship etc with patient and family
- Genetic testing in certain disorders but needs specific genetic counsellor involvement

Epilepsy and Psychiatry

Epilepsy can be associated with psychiatric disorders in the following ways

- Psychiatric disorders caused by the seizures of the epilepsy; Ictal disorders, postictal disorders, and interictal disorders
- Epileptic and psychiatric disorders caused by common brain pathology
- Epileptic and psychiatric disorders that happen to coexist in the same patient but are not causally related.

Epilepsy and Psychosis

- 2–9% of patients with epilepsy have psychotic disorders

Ictal Psychosis

- Status epilepticus (i.e., complex partial status epilepticus and absence status epilepticus) can mimic psychiatric disorders, including psychosis.

Post-ictal Psychosis

- Psychotic or other psychiatric symptoms occur after a seizure or, more frequently, a series of seizures, after a lucid interval or within 7 days of the seizure(s).
- The event may be psychosis, depression or elation, or an anxiety-related symptom.
- The event is time limited, lasting days and rarely weeks. No significant clouding of consciousness occurs.
- Logsdail and Toone believe that clouding of consciousness, disorientation, or delirium may be noted and, if consciousness is unimpaired, delusions and hallucinations are present; a mixture of both also may be noted.

Inter-ictal Psychosis

- Tandon and DeQuardo reviewed the series of patients with epilepsy who developed psychosis and found that the psychosis was usually a form of schizophrenia, most commonly paranoid schizophrenia.
- Stagno reported that persistent inter-ictal psychoses of epilepsy and the “schizophrenia-like psychoses of epilepsy” are distinguishable from schizophrenia in the traditional psychiatric sense by the following
 - Lack of negative symptoms of schizophrenia, particularly flattening of affect and personality deterioration
 - Better premorbid personality
 - Paranoid delusions, delusions of reference
 - More benign and variable course

Management

- Ictal psychosis and post-ictal psychosis are treated by optimising anticonvulsant drugs. Post-ictal psychosis is self-resolving, but antipsychotic drugs may hasten its resolution
- Chronic inter-ictal psychosis is treated with antipsychotics
- Most antipsychotics can reduce the seizure threshold and the risk is dose related
- Clozapine is most epileptogenic: 4–5% of those who receive doses more than 600 mg have seizures
- Sodium valproate is the anticonvulsant of choice for Clozapine-induced seizures.
- Antipsychotics with low epileptogenic potential include older antipsychotics like Haloperidol
- Sulpiride is also a good choice

Depression in Epilepsy

- 9–22% of epileptics suffer clinical depression
- Suicide rates 4–5 times that of normal population
- Moclobemide and SSRIs good choice
- Mirtazapine and Venlafaxine – with caution
- Most anticonvulsants can cause depression
- Fluoxetine and Paroxetine are potent inhibitors. Phenytoin and Carbamazepine levels can be increased
- Antiepileptics can decrease folate levels which can contribute to depression.

The etiology of depression in people with epilepsy is complex and includes the following:

- Genetic vulnerability
- Reaction to life events, including the epilepsy
- Effect of antiepileptic drugs

IMPORTANT DRUG INTERACTIONS TO REMEMBER IN EPILEPSY

- **Psychotic symptoms:** Carbamazepine, ethosuximide, tiagabine, topiramate, valproate and vigabatrin are associated with psychosis
- Case reports of psychosis with **levatiracetam (Kepra)**
- All anticonvulsants are associated with **depressive symptoms**
- **Pharmacokinetic interactions** between anticonvulsants, antidepressants and antipsychotics are common
- **Fluoxetine and paroxetine** are potent inhibitors of several hepatic enzymes (CYP2D6, CYP3A4)
- **Carbamazepine and phenytoin** have low therapeutic indexes and their levels can be increased by enzyme inhibitors
- **Carbamazepine** is an enzyme inducer (CYP3A4) and can lower levels of antipsychotic drugs

Parkinson's Disease

- Loss of pigmented dopaminergic neurons in the substantia nigra and presence of Lewy bodies
- Lewy bodies are concentric, eosinophilic, cytoplasmic inclusions with peripheral halos and dense cores
- **Clinical signs**
 Asymmetric resting tremor (cf. EPSE- symmetrical), 3–5 Hz. Pill rolling type
 Bradykinesia expressed as micrographia, hypomimia, hypophonia and decreased blink rate
 Rigidity – cog wheel and lead pipe
 Gait – shuffling, stooping

Management

- L-DOPA + Carbidopa- metabolic precursor of dopamine + peripheral decarboxylase inhibitor
- Anticholinergic drugs
- Entacapone- increases half-life of L-DOPA- associated with less motor fluctuations and less dyskinesia
- Dopamine agonists- bromocriptine, amantadine, pergolide, pramipexole
- Selegline- MAO-B inhibitor. Caution if concurrent SSRI administration

IMPORTANT POINTS IN MANAGEMENT OF DEPRESSION IN PARKINSON'S DISEASE

- Exclude organic cause such as hypothyroidism – high prevalence in Parkinson's disease
- SSRIs first line, but caution when used with selegline as can cause serotonin syndrome
- Avoid TCAs as can cause constipation, confusion, and can worsen autonomic dysfunction

- Consider ECT – depression and motor symptoms respond well but risk of delirium is high
- Consider augmentation with dopamine agonists like pramipexole or amantadine
- If no response, treat as treatment resistant depression

IMPORTANT POINTS IN MANAGEMENT OF PSYCHOSIS IN PARKINSON'S DISEASE

- Exclude organic causes
- If patient has psychotic symptoms such as visual hallucinations that are not distressing treatment may not be required
- Reduce or stop anticholinergics and dopamine agonists
- Antipsychotics can worsen movement disorder and dopamine agonists can worsen psychosis, therefore an optimal balance is necessary
- Consider atypical antipsychotic such as quetiapine
- Consider a cholinesterase inhibitor particularly if the patient has co-morbid dementia
- Try clozapine in a small dose, up to 25 mg/day
- Consider ECT

Management of Depression in Physical illness/Organic Case

- Consider the setting; medical or psychiatric ward
- Assess severity of depression
- Assess risk
- Multidisciplinary team involvement; doctors, nurses, occupational therapists, physiotherapists, pharmacist and psychologists
- Psychological treatment; CBT, IPT may be used as adjuncts
- Medication
 - **Lithium and NSAIDs**- increased lithium levels
 - Reduce **propranolol** dose if SSRIs used
 - Reduce dose of TCA or Moclobemide if on **cimetidine** (enzyme inhibitor)
 - Increase dose of TCA if on high fibre diet
 - Reduce and monitor **warfarin** if on fluoxetine or fluvoxamine
 - Reduce dose of **donepezil** if on SSRI
 - Avoid SSRI and MAOI with opioids
 - Avoid TCA and carbamazepine if receiving chemotherapy
 - Caution when using SSRI with **selegiline** for Parkinson's disease

Cardiovascular disease

- SSRI first choice
- Mirtazapine safe in arrhythmias
- Lithium can cause ECG changes

Hepatic disease

- Low dose paroxetine or citalopram
- Low dose imipramine
- Avoid Lofepramine, nefazodone, and sertraline
- Avoid TCAs – can mask hepatic encephalopathy
- Avoid MAOIs – hepatotoxic

Renal disease

- No clear choice but citalopram and sertraline are reasonable choices

Diabetes mellitus

- SSRIs may reduce serum glucose
- TCAs to be avoided – may increase serum glucose levels by 150%
- Amitriptyline, imipramine and citalopram can be used to treat diabetic neuropathy

Post-Stroke Depression

- SSRI or nortriptyline

Parkinson's disease

- TCA's good for anticholinergic effects
- SSRIs – caution with selegiline (risk of serotonin syndrome)
- Lithium and valproate may exacerbate tremor

PSYCHIATRIC DISORDERS IN PREGNANCY AND POSTPARTUM

PUERPERAL PSYCHOSIS

Risk factors for puerperal psychosis

- Past history of puerperal psychosis (50–90%)
- Past history of bipolar disorder (20–30%)
- Family history of bipolar affective disorder (50%)

▲ IMPORTANT POINTS IN MANAGEMENT

1. Risk

- Evaluate risk of harm to self or baby
- Suicidal ideation
- Homicidal ideation
- Delusions related to baby
- Social support
- Presence of partner
- Decide on in-patient or out-patient management and necessity of Mental Health Act
- Mother-baby unit
- Supervision levels

2. Clarification of diagnosis

- Collateral
- GP
- Family
- Previous notes
- Response to medication
- Investigations
- FBC, U&E, LFT – electrolyte imbalance and infections are common postpartum
- Blood levels – lithium, valproate etc.
- B12, folate
- TFT can exacerbate depression

3. Treatment of acute psychological symptoms

Choice of drug will depend on whether mother is breastfeeding

General principles of prescribing psychotropics in breastfeeding

- Treatment of illness is highest priority
- Risk-benefit analysis
- Premature infants and infants with renal, hepatic, cardiac or neurological impairment are at greater risk from exposure to drugs.
- Monitor the infant closely for abnormal cries, failure to thrive, abnormal feeding and sleeping patterns and growth and development
- Use the lowest possible dose
- Avoid polypharmacy

- Time feeds to avoid peak levels or express milk
- Mothers taking psychotropics during pregnancy are to be monitored in a mother-baby unit for three days postpartum for withdrawal symptoms in the baby
- Antipsychotics in breast feeding; Sulpiride / olanzapine
- Antidepressants – paroxetine (lowest milk/plasma ratio and shortest half-life) or sertraline. Avoid doxepin
- Mood stabilisers – avoid if possible, valproate if essential
- Sedatives – lorazepam for anxiety, zolpidem for sleep
- Supportive counselling
- Once acute situation is resolved, use psychoeducation of patient and family
- Referral and liaison with neonatology, obstetrician and GP
- Discharge with acute follow up
- Relapse prevention – compliance therapy
- Drug and alcohol counselling if necessary
- Discuss implications of psychosis and medications if planning for further pregnancies and need to plan

General principles in prescribing in pregnancy

- Discuss possibility of pregnancy
- Importance of planning pregnancies and use of contraception
- Psychoeducation is crucial
- Risk-benefit analysis for each case. - discontinuation of medication abruptly post conception for women with severe mental illness and those at high rate of relapse is not recommended
- Perinatal services referral
- Involve specialist perinatal psychiatrists if possible
- Avoid drugs in first trimester, use lowest effective dose
- Avoid polypharmacy
- Foetal screening
- Consider changes in body volume resulting in plasma level fluctuations in pregnancy. Drug doses may need to be adjusted, particularly in third trimester.
- Drug dose may be tapered in third trimester depending on patient's history and following discussion with patient and her partner.

DRUGS IN PREGNANCY – IMPORTANT POINTS

- Antipsychotics; Olanzapine and older antipsychotics
- Recent evidence also shows quetiapine is safe
- Antidepressants; Amitriptyline, imipramine and fluoxetine
Antidepressants are linked with reduced birth weight and earlier delivery
- Avoid paroxetine – linked with persistent pulmonary hypertension of the newborn (PPHN) especially with third trimester administration
- Monitor the infant after delivery. Most mother-baby units will have a minimum of three days after birth to observe for signs of withdrawal; irritability, crying, sluggish reflexes, fever, poor appetite, failure to thrive and convulsions. Risk is greater with short half-life drugs such as paroxetine and venlafaxine
- ECT is effective for severe psychotic depression

Preventative strategies in postpartum disorders

Prevention is always divided into:

Primary– aimed at reducing the prevalence of condition

Secondary – aims to prevent the further progression of a disease or condition by identifying and treating cases at an early stage

Tertiary– preventing condition from occurring after it has occurred once

PRIMARY PREVENTION

- Educational programmes about detection and early treatment – media, GP clinics, community talks
- Folate and vitamin supplementation
- Diet and nutritional advice

SECONDARY PREVENTION

- Antenatal screening
- Intensive monitoring during early postpartum period
- Edinburgh Postnatal Depression Scale (EPDS) (a 10-item self-report scale) administered two or three times by health visitors during first 6–8 months

TERTIARY PREVENTION

- Relapse prevention strategies as discussed above
- Antidepressant, mood stabiliser, antipsychotic prophylaxis
- Social support

POSTNATAL DEPRESSION

Risk factors for Postnatal Depression

Based on meta-analysis by O'Hara and Swain

- Past history of depression
- Psychological issues in pregnancy
- Poor social support and relationship with partner
- Recent adverse life events
- Less well associated factors include:
 - Obstetric complications
 - Lower socio-economic status
 - History of abuse
 - Severe postnatal “blues”

Consequences of postnatal depression for child

- Learning problems
- Language difficulties
- Cognitive impairment esp. boys (Murray & Cooper XXXX)
- Impaired bonding
- Impaired social and emotional development

IMPORTANT POINTS IN BIPOLAR DISORDER AND PREGNANCY

Avoid mood stabilisers in pregnancy unless the risks of not treating outweigh the risks of teratogenesis (NICE Guidelines UK)

Lithium

- Lithium is associated with Ebstein's anomaly (RR- 10–20 more than control but absolute risk is low – 1:1000)
- Maximum risk in first trimester esp. first 2–6 weeks
- If lithium is continued, high resolution ultrasound and echocardiography should be performed at 6 and 18wks of gestation
- In the third trimester – increasing dose of lithium is required as total body water increases
- Levels may rise abruptly after delivery

Valproate and carbamazepine

- Causal link with spina bifida, with valproate having a greater risk
- If continued therapy is required, then low dose monotherapy in conjunction with high dose folate (5mg /day)
- Ideally all women should take folic acid for at least one month before conception (may reduce risk of neural tube defects)
- Administer prophylactic vitamin K to mother and neonate after delivery

Lamotrigine

- Associated with cleft palate

EATING DISORDERS

IMPORTANT POINTS IN FORMULATION AND MANAGEMENT

- Personal history is important to establish risk factors below
- Family history will reveal history of weight dysfunction or eating rituals
- Psychodynamic model; use of eating or starvation as only situation that individual is able to control in an invalidating environment
- Cognitive model – more applicable to bulimia but can be modified to anorexia
- Bingeing or starvation as a coping mechanism
- Physical examination is a must
- Physical risk is high
- Consider setting of treatment
- Emergency medical treatment
- Liaise with medical team and dietician input
- Beware re-feeding syndrome
- Relapse prevention is based on addressing stressors/psychoeducation and CBT or family therapy

ANOREXIA NERVOSA

Risk factors for anorexia nervosa

- Family history of eating disorders, parental obesity, restrictive dieting, concerns about eating, appearance or weight
- Personal H/o-Body dissatisfaction, restrictive dieting, childhood obesity, early menarche (<12 yrs.), depression, substance abuse or dependence, adverse life events
- Personal characteristics: perfectionist or obsessional traits, alexithymia, low self esteem
- Female and occupational status that emphasises appearance

THE MAIN ISSUES

1. Risk

- Physical complications (see later for health risks of anorexia nervosa)
- Site for treatment

2. Treatment of condition

- Restore patient to healthy weight (menses in females, normal sex drive in males, pubertal development in children and adolescents)
- Psychoeducation regarding disorder, and healthy eating and nutrition
- Cognitive assessment and addressing negative cognitions
- Treat co-morbidity
- Family psychoeducation and therapy
- CBT
- Pharmacological treatment

3. Clarification of diagnosis

- Assessment of co-morbidities
- Metabolic investigations
- Physical examination
- Collateral
- Relapse prevention

The aims of treating anorexia nervosa are to

- **Restore patients to a healthy weight** (associated with the return of menses and normal ovulation in female patients, normal sexual drive and hormone levels in male patients, and normal physical and sexual growth and development in children and adolescents)
- **Treat physical complications**
- **Enhance patient's motivation** to cooperate in the restoration of healthy eating patterns and participate in treatment
- **Provide education** regarding healthy nutrition and eating patterns
- Help patient **reassess and change core dysfunctional cognitions**, attitudes, motives, conflicts, and feelings related to the eating disorder
- **Treat associated psychiatric conditions**, including deficits in mood and impulse regulation and self-esteem and behavioural problems
- **Enlist family support** and provide family counselling and therapy where appropriate
- **Prevent relapse.**

HEALTH RISKS OF ANOREXIA NERVOSA

Cardiovascular complications

- Bradycardia
- Hypotension
- Sudden death
- Mitral valve dysfunction
- Arrhythmias (more common with bingeing and purging)
- Cardiac failure (more common with bingeing and purging)

Electrolyte complications

Bingeing/ Purging/Vomiting

- Hypokalaemia
- Hyponatraemia
- Hypochloraemia
- Metabolic alkalosis
- Metabolic acidosis (laxative abuse)
- Hypomagnesaemia, hypophosphatemia
- Hypocalcaemia

Endocrine and metabolic abnormalities in eating disorders

- Endocrine abnormalities occur mainly in Anorexia Nervosa
- Amenorrhea
- Low LHRH, LH, FSH
- Low oestrogen and progesterone
- Low T3
- High cortisol
- High fasting GH
- Hypercholesterolemia
- Hypercarotenaemia
- Impaired glucose tolerance

Haematological complications in eating disorders

- Leukopenia
- Lymphocytosis
- Thrombocytopenia
- Low ESR
- Reduced serum complements levels

GI complications

- Parotid swelling
- Refeeding pancreatitis
- Delayed gastric emptying
- Nutritional hepatitis
- Constipation

Purging and vomiting

- Oesophageal erosion
- Gastric and duodenal ulcers
- Pancreatitis

Indications for in-patient admission in anorexia nervosa

- Medical status: Adults HR<40/min, BP 90/60mmhg, low potassium, Temp <97, >20bpm HR increase, >20mmHg drop in BP
- Active suicidal plan
- BMI <14
- Rapid weight loss
- Continuous preoccupation with cognitions
- Needs supervision at mealtimes
- Naso-gastric feeding
- Exercise needs restraint
- Compulsive uncontrolled purging
- Severe family dysfunction
- Psychiatric disorder requiring treatment

Based on risk patient management takes place in the following settings

- In-patient settings
- Residential
- Partial hospitalization
- Out-Patient

TREATMENT FOR ANOREXIA NERVOSA

1. Establish therapeutic alliance
 2. Treat as part of a multidisciplinary team – psychologists, endocrinologists, dieticians, dentists, psychiatrists, social worker, and occupational therapists.
 3. Provision of educational materials, including self-help workbooks; information on community-based and Internet resources; and direct advice to patients and their families (if they are involved). A team approach is the recommended model of care.
 4. Assess and monitor eating disorder symptoms and behaviours
 5. Legal interventions, including involuntary hospitalization and legal guardianship, may be necessary to address the safety of patients who are reluctant to receive treatment but whose general medical conditions are life threatening
 6. Nutritional rehabilitation
 - 2–3 lb/wk. for in-patients / 0.5–1lb/wk. out-patients
 - Structured meal plan by dietician to ensure nutrition across broad range of foods
- Caloric intake – 30–40 kcal/kg/day up to 70–100 kcal/kg/day in some patients who may require a large quantity of calories to gain weight. If nasogastric feeding, continuous feeding (preferable to IV), 24 hrs/day causes less metabolic abnormalities.
 - Weigh patients at fixed time during day in same attire after voiding
 - Meals eaten together
 - Bed rest after meals
 - Physical activity adapted to food intake and energy expenditure
 - Once safe weight achieved, focus on physical fitness rather than expending calories
 - Staff to deal with concerns about body weight and image changes
 - Considerable evidence that as weight is restored, symptoms of eating disorder abate
 - This is usually associated with improvement in mood and anxiety symptoms

Refeeding syndrome

- Occurs with severely malnourished patients
- Hypophosphatemia, hypomagnesemia, hypocalcaemia and fluid retention
- **Clinical symptoms:** dyspnoea, generalised weakness, fatigue, peripheral oedema, seizures and coma, paraesthesias and auditory hallucinations and sudden death
- **Treatment:** bed rest, IV fluids, multivitamins (thiamine supplements, sodium phosphate, zinc gluconate and potassium chloride)
Suspend nutrient intake or reduce significantly.
- **Prevention:** Monitor serum levels of PO₄, Mg, and Ca for first five days of refeeding and every other day for several weeks.

7. Psychosocial interventions

Acute Anorexia Nervosa

- Empathy, explanations, praise, encouragement and other positive reinforcement
- Children and adolescents – Family treatment is most effective intervention

After weight restoration

- Psychoeducation about illness
- CBT has good evidence; corrects erroneous beliefs about food and weight, bolsters motivation, and improves self-esteem.
- IPT also shows some evidence
- CBT helpful in preventing relapse - RCT
- For adolescents and children – family therapy.
- Anorexics Anonymous and Bulimics Anonymous – not a substitute for treatment. Not adequately studied

Chronic Anorexia nervosa

- Lack of response to psychotherapy
- But literature does show cases that have spontaneously remitted.

8. Pharmacological treatment of anorexia nervosa

- SSRI: useful for depressive, anxiety or obsessive-compulsive and bulimic symptoms in weight restored patients
- Fluoxetine 60 mg/day: decreases relapse episodes in weight restored patients but for patients receiving CBT after weight restoration, adding Fluoxetine does not confer additional benefits
- Antipsychotics: SGAs may be beneficial during the weight restoration phase or for marked obsessionality, anxiety, limited insight and hyperactivity
- No good evidence for other medications, e.g. zinc, cisapride

9. Relapse Prevention

- Evidence that fluoxetine at 60 mg has some benefit
- CBT

TREATMENT FOR BULIMIA NERVOSA

The aims of treatment for patients with bulimia nervosa are to

- Reduce and, where possible, eliminate binge eating and purging
- Treat physical complications of bulimia nervosa
- Enhance patient's motivation to cooperate in the restoration of healthy eating patterns and participate in treatment
- Provide education regarding healthy nutrition and eating patterns; help patient reassess and change core dysfunctional thoughts, attitudes, motives, conflicts, and feelings related to the eating disorder
- Treat associated psychiatric conditions, including deficits in mood and impulse regulation, self-esteem, and behaviour

- Enlist family support and provide family counselling and therapy where appropriate; and prevent relapse.

Nutritional rehabilitation counselling

- Help the patient develop a structured meal plan as a means of reducing the episodes of dietary restriction and the urges to binge and purge
- Adequate nutritional intake can prevent craving and promote satiety.
- It is important to assess nutritional intake for all patients, even those with a normal body weight (or normal BMI), as normal weight does not ensure appropriate nutritional intake or normal body composition
- Nutritional counselling is a useful part of treatment and helps reduce food restriction, increase the variety of foods eaten, and promote healthy but not compulsive exercise patterns

Psychosocial interventions

- Initial treatment – Self-help programmes and professionally guided self-help programmes
- CBT – the most efficacious
- Education about disorder
- Addressing cognitive distortions related to food, weight and self
- Link between dietary restriction, binge eating and purging
- If no initial response to CBT – IPT or high-dose fluoxetine useful in some cases
- Family and other group psychotherapies have some utility
- Family therapy should be considered for children and adolescents
- Long-term support group

Pharmacological management of Bulimia nervosa

- High-dose Fluoxetine 60 mg/day is effective in treating bulimic symptoms
- Also useful in depressive, anxiety, obsessive-compulsive symptoms or other impulse disorder symptoms
- Small controlled trials have shown benefit of Topiramate. Weight loss is a side effect
- Combination of antidepressant and CBT resulted in highest remission rates (2 RCTs)
- Ondansetron 5HT₃ antagonist in unusual circumstances.
- Bright light therapy – not conclusive
- rTMS – not conclusive

DRUG AND ALCOHOL MISUSE AND DEPENDENCE

ALCOHOL

ALCOHOL WITHDRAWAL

Supportive care

- Environment – minimise stress by making sure area is bright, quiet, safe and private
- Psychoeducation – Information to patient about withdrawal, course, risks associated
- Reassurance – Most effective intervention in reducing severity of withdrawal symptoms.
- Coping skills- – relaxation management, mindfulness, dietary guidelines, sleep, hygiene, methods to reduce craving

Monitoring

- Clinical Institute of Alcohol Withdrawal Scale (Revised) - CIWA-Ar
- AWS
- TPR, BP
- Level of hydration – urine output
- Inform if AWS>15 or CIWA>20
- Bloods; LFT, GGT, U and E, FBC, CRP, CXR

Treatment

- Diazepam sliding scale; 5–20 mg up to maximum of 80–120 mg according to AWS or CIWA
- Prevention of dehydration – oral fluids/ IV fluids
- Prevention of Wernicke's encephalopathy
- 100 mg TDS orally for one week, or if severe 200–300 mg IM daily for 3–5 days
- If Wernicke's encephalopathy confirmed; two pairs of IV ampoules of pabrinex (high potency B complex vitamins) TDS for two days
- If liver disease use oxazepam or lorazepam
- Psychosis or hallucinations – oral haloperidol or small dose of typical antipsychotic
- Treat symptomatically – diarrhoea- fluid balance; pain- paracetamol
- Treatment of delirium tremens (see later)

Discharge planning once acute withdrawal stage treated

- Relapse prevention strategies (see later)
- Social factors; extent of social network, accommodation, employment
- Drug and alcohol services

CLINICAL FEATURES OF WERNICKE'S ENCEPHALOPATHY

- Ataxia
- Confusion
- Ophthalmoplegia / nystagmus (abducens or sixth nerve palsy)
- Hypothermia or hypotension
- Memory disturbance

- Coma/Unconsciousness

FEATURES OF KORSAKOFF SYNDROME

- Severe retrograde and anterograde amnesia particularly anterograde
- Telescoping of events
- Confabulation
- Polyneuropathy
- Lack of insight
- Apathy
- Deficiency of thiamine
- Necrosis and haemorrhage in the mamillary bodies

CLINICAL FEATURES OF DELIRIUM TREMENS

- Medical emergency
- Usually occurs in 48–72 hrs after abstinence from alcohol but may occur up to seven days later
- Exaggerated features of simple withdrawal in 75% of cases, i.e., anxiety, dehydration, sweating, tachycardia, headaches, insomnia and tremor.
- Autonomic instability
- Hyperthermia
- Seizures
- Coarse tremor
- Disorientation and confusion
- Hallucinations- Lilliputian type (EMQ)
- Paranoid delusions
- Complications; arrhythmias, HT, pneumonia, renal failure

RELAPSE PREVENTION

Evidence based interventions for Alcohol Dependence (Advances in Psychiatric treatment, APA guidelines)

PSYCHOLOGICAL AND SOCIAL TREATMENTS FOR ALCOHOL DEPENDENCE

- Alcoholics Anonymous
- Motivational enhancement therapy
- CBT (Marlatt & Gordon)
- Social skills training; larger repertoire of coping skills will reduce the stress of high-risk situations and provide alternatives to alcohol use. Techniques involve assertiveness training, modelling, and role playing of skills such as refusal of alcohol and dealing with interpersonal problems.
- Social behaviour and network therapy (UKATT study); Combination of cognitive behavioural approach and community reinforcement approach, behavioural marital and family therapy.
- Contingency management:
 - Clinician arranges the environment such that alcohol use is readily detectable

Reinforcers are arranged to reward abstinence

Incentives are withheld following alcohol use

Reinforcement from alternative sources (employment, family or social) is increased to compete with that from alcohol

- Cue exposure – repeated exposure to stimuli to extinguish cravings
- Therapeutic communities and residential rehabilitation

BIOLOGICAL TREATMENTS FOR ALCOHOL DEPENDENCE

- **Disulfiram**
 - a. Prevents breakdown of alcohol by aldehyde dehydrogenase
 - b. Can cause liver dysfunction and thyroid dysfunction
 - c. Principle of negative reinforcement
- **Naltrexone** - Opiate receptor antagonist that reduces the pleasurable effects of drinking
- **Acamprosate** - GABA analogue

OPIOIDS

OPIOID DEPENDENCE

Biological treatments

- **Methadone** maintenance. Opioid agonist, 20–100 mg oral
- **Buprenorphine**- Partial Agonist – 8–24 mg sublingually
- **Naltrexone**- Opioid antagonist
- Clonidine and Lofexidine. Alpha-2 adrenergic agonist
- LAAM- Longer acting opioid agonist than methadone
- Recently, Buprenorphine-naloxone combination (Suboxone) as a long acting tablet. Aim is to prevent diversion and facilitate unsupervised dosing programmes

Psychological treatments

- Psychoeducation – Identify harmful behaviours
 - Needle sharing
 - Where do they buy heroin from?
 - Variation in purity
 - Injection sites
- Education about risks of overdose, risk of transmission of illness, managing overdose, medical illness and contraception
- Counselling about less risky practices and needle exchange services
- Screening for HIV, Hepatitis B and C
- Encourage vaccination for all patients
- CBT
- Contingency management
- Psychodynamic and Interpersonal therapies- Psychodynamically oriented group therapy, modified for substance-dependent patients, appears to be effective in

promoting abstinence when combined with behavioural monitoring and individual supportive psychotherapy. The role of IPT is unclear,

- Family therapy – enhance treatment adherence and facilitate implementation and monitoring of contingency contracts with opioid-dependent patients
- Twelve-step programmes

FEATURES OF OPIOID WITHDRAWAL

- Lactorrhoea
- Rhinorrhoea
- Agitation
- Perspiration
- Piloerection
- Pulse rate (BPM)
- Vomiting
- Shivering
- Yawning /10 min
- Dilated pupils (cf. pinpoint pupils in overdose)
- Clinical institute Narcotic assessment (Withdrawal scale)

MANAGEMENT OF OPIOID WITHDRAWAL

- Methadone substitution, with gradual methadone tapering
- Abrupt discontinuation of opioids, with use of clonidine to suppress withdrawal symptoms
- Clonidine-naltrexone detoxification, where withdrawal symptoms are precipitated by naltrexone and then suppressed by clonidine
- Buprenorphine substitution, followed by abrupt or gradual discontinuation of buprenorphine; and
- Use of other medications to treat the symptoms of opioid withdrawal.

PRINCIPLES IN METHADONE TREATMENT

- Confirm dependence and opioid use; history, physical examination, urine tests and corroborative sources
- **Obtain informed consent** from patient prior to commencement
- Nature of treatment
- Purpose of treatment
- Side effects
- Risks associated with administration
- Benefits
- Risks associated with not having
- Can withdraw consent at any time
- Programme policies and expectations
- Leaflets and drug and alcohol team involvement

- If patient already on methadone following conditions to be met before administration of dose:
 - I. Confirm dose with prescriber
 - II. Last consumption confirmed and supervised
 - III. Prescriptions not being issued elsewhere
 - IV. Patient comfortable on dose
 - V. No other CI present
- **LFTs and ECG-** risk of hepatic or renal dysfunction can affect excretion and methadone can prolong QT interval
- Start low, 5–10 mg for moderate withdrawal. Remember that doses > 30 mg have a high potential for overdose and death if opioid naive
- Further incremental doses can be given in 5–10 mg increments according to withdrawal scale
- Always prescribe under supervision
- Stabilisation usually achieved within 6 weeks. And between 60–120 mg/day
- Overdose- naloxone IV- 0.4–2 mg. Effect can reverse in 20 min to 1 hr. so monitor and repeat administration.

PRINCIPLES IN MANAGEMENT OF BENZODIAZEPINE DEPENDENCE

- Ensure that person has no indication for which they need it
- Treat co-morbid anxiety or depression with appropriate medication e.g. antidepressants
- Psychoeducation about benzodiazepines and dependence potential
- Motivational interviewing to establish readiness to change
- Change to long-acting BZD if on short-acting BZD
- Gradual reduction of BZD dose, 10% every week / fortnight
- Monitor for seizure activity and anxiety
- Relapse prevention after detoxification period has elapsed

SMOKING

Physical conditions associated with smoking

- CHD
- Peripheral vascular disease
- Chronic bronchitis
- 80% of lung cancers
- Thromboembolic disease
- Nasopharyngeal and upper GI carcinoma

General features to encourage smoking cessation (4As)

- Ask about smoking at least once a year
- Advise smokers to stop and refer to specialist services
- Assist smokers to stop

- **Arrange follow up**

SMOKING CESSATION INTERVENTIONS

1. NRT

- gum, patch, inhaler, nasal spray, tablet and lozenge
- Combining different forms of NRT enhances cessation rates
- Pre-loading with NRT may increase cessation rates
- Use of NRT alongside a reduction in cigarette consumption may be useful as a first step towards cessation for heavier dependent smokers

2. Bupropion

- Dopamine and norepinephrine reuptake inhibitor.
- Reduces withdrawal symptoms and post cessation weight gain.
- Side effects; psychiatric disturbances, seizures

3. NRT+Bupropion – doubles cessation rates

4. Varenicline

- Nicotine partial receptor agonist. Significantly higher cessation rates compared to placebo and bupropion (1.6 times)
- Side effects; nausea, dry mouth, psychiatric disturbances, case report of manic episode, depression and suicidal ideation

5. Nortriptyline – effective

6. Clozapine enhances smoking cessation in schizophrenia

7. Brief advice from GP; increases abstinence rates at 6 months by 1–3% above controls

8. Specialist clinics – 17% above controls

Effects of smoking on psychiatric medications

- CYP1A2- olanzapine, clozapine, diazepam and TCA's. Smoking can lower the levels of these drugs.
- Clozapine levels have been found to be 40% lower in smokers than in non-smokers

GERIATRIC PSYCHIATRY

FORMULATION MODELS

- Biological models – medications causing depression, cardiovascular risk factors
- Vascular depression (biological model)
- Cognitive models – cognitive therapy however may need to be modified to adapt to age
- Erikson's psychosocial models
- Hidebrand's psychosocial models (see exam resources power point slides)

Dementia

Common types of Dementia:

- Alzheimer's disease (60% of cases)
- Vascular dementia (30–40%; including about 20% where dual pathology exists)
- Dementia with Lewy bodies (15% of cases)
- Fronto-temporal dementia (5%)
- Main differences between sub-cortical dementia and cortical dementia are that speech and memory are not affected in sub-cortical but speed of movement, mood changes, personality changes and frontal and visuo-spatial functions are affected in comparison to cortical dementia in which speech, memory and visuo-spatial functions are mainly affected and early in the dementia

DSM –IV CRITERIA FOR ALZHEIMER'S DISEASE

- Memory impairment manifested by impaired ability to learn new information (anterograde amnesia) or to recall previously learnt information (retrograde amnesia)
- Aphasia
- Apraxia
- Agnosia
- Disturbance in executive functioning (planning, organizing, sequencing, abstracting)

RISK FACTORS FOR ALZHEIMER'S DISEASE

- Age
- Family history- 1 in 5 and 1 in 6 persons who are relatives of patients with Alzheimer's disease will develop the disease
- Down syndrome
- ApoE 4- the E4E4 genotype is associated with much higher risk of AD compared with E3E3 genotype. The E3 allele is less common in Orientals than in Caucasians, which might account for lower incidence of Alzheimer's in Orientals
- Age of onset before age of 60–65
- AD inheritance for 5% cases
- Three genes account for 30–50% of familial Alzheimer's disease

DIFFERENCES BETWEEN EARLY ONSET AND LATE ONSET ALZHEIMER'S DISEASE

- Age of onset before age of 60–65
- AD inheritance for 5% cases
- Three genes account for 30–50% of familial AD
- Genes associated with early onset Alzheimer's disease:
 - APP gene (Beta-Amyloid precursor protein) located on chromosome 21
 - Autosomal dominant pattern of inheritance
 - Presenilin genes
 - PS-1 located on chromosome 14
 - PS-2 located on chromosome 1
 - Autosomal dominant penetration

CHARACTERISTICS OF FRONTO-TEMPORAL DEMENTIA

- Males > females
- Age 45–65
- Positive family history in 50% of patients
- Association between fronto-temporal dementia and motor neuron disease

Behavioural features

- Personality change – coarsening of personality
- Mental rigidity and inflexibility
- Perseverative and stereotyped behaviour
- Utilization behaviour
- Reduced motivation

Speech and Language

- Loss of verbal fluency
- Perseveration
- Stereotypy
- Echolalia

Physical signs

- Primitive reflexes
- Akinesia
- Rigidity
- Tremor
- Incontinence

The above features are also present in sub-cortical dementias – e.g. HIV, Parkinson's disease, Huntington's disease.

CRITERIA FOR LEWY BODY DEMENTIA – IMPAIRED COGNITION CONSISTENT WITH DEMENTIA PLUS

- Fluctuating cognition with pronounced variation in alertness and attention
- Recurrent visual hallucinations that are well formed and detailed
- Spontaneous features of parkinsonism
- Supportive features
- REM sleep behaviour disorder
- Severe neuroleptic sensitivity
- Low dopamine transporter uptake in basal ganglia demonstrated by SPECT or PET imaging
- Falls and syncopal episodes in about 1 in 3 patients

MANAGEMENT OF DEMENTIA

Multimodal and multidisciplinary treatment

Risk

- Physical risk
- Neglect
- Risk of harm to self: suicide and accidental self-harm
- Risk of wandering
- Risk of falls
- Financial risk
- Risk of exploitation

Alliance with family and patient

Determine the site of treatment – home treatment with carer, in-patient or high level or low-level nursing home

Diagnostic evaluation

- FBC, U&E, LFT, B12 and Folate, TFT, Syphilis serology, glucose, cholesterol
- ECG, EEG, CXR, Autoimmune screen
- Electrolyte abnormalities can cause delirium
- LFTs to check liver function as medications if necessary are metabolised through the liver
- Renal function tests – excretion of drugs and caution with lithium
- Lipids and glucose – antipsychotics and possibility of vascular risk factors
- Neuroimaging
 - MRI
 - PET- differentiating between fronto-temporal dementia and Alzheimer's dementia
 - Biomarkers- CSF for CJD

Neuropsychological testing

- Extent of dementia
- Distinguish between dementias
- Establish baseline cognitive function

- Identifies strengths and weaknesses
- Genetic testing not recommended
- Refer to genetic counsellor

Rating scales

ADAS-COG, MMSE, geriatric depression scale, HoNOS 65+

Cohen-Mansfield Agitation inventory

- Manage risks
- Management of behavioural and psychological symptoms of dementia
- Treatment of dementia with cholinesterase inhibitors if indicated
- Education and support to the family
- Refer family to appropriate sources of care
- Assessment of activities of daily living – Occupational therapist
- Carer's assessment – social worker
- Counselling
- Support groups (Alzheimer's Association)
- Financial help, carer's allowance
- Meals on wheels
- Home care
- Day hospital
- Transport to attend groups
- Financial planning
- Geriatric law specialists

Watch for signs of distress in caregiver

- Assessment of carer
- Carers support groups
- Causes for caregiver distress
- Care recipient's behaviour
- Care duties
- Carer's fears and worries

Support families during decision about institutionalisation – Address Legal and financial issues

- Power of Attorney
- Testamentary capacity
- Advance directives
- Guardianship
- Financial planning – finances required for placement

BEHAVIOURAL AND PSYCHOLOGICAL SYMPTOMS OF DEMENTIA

- Encompasses all symptoms of disturbed perception, mood, thought and behaviour that occur in patients with dementia (Finkel & Burns 2000)
- Forgetfulness
- Disorientation
- Wandering
- Incontinence
- Aggression
- Sexual disinhibition
- Delusions
- Uncontrolled emotions
- Insomnia

PRINCIPLES IN THE MANAGEMENT OF BEHAVIOURAL AND PSYCHOLOGICAL SYMPTOMS OF DEMENTIA (BPSD)

GENERAL PRINCIPLES

- Exclude physical cause
- Behavioural management first
- Maintain consistency
- Preserve dignity
- Encourage normal behaviour
- Avoid punishment

Physical environment

- Familiar constant and stress free
- Careful design and good staffing
- Buildings for people should
 - Make sense, help them find their way
 - Provide a therapeutic environment
 - Provide a safe environment
 - Provide good facilities for staff

Sensory functions

- Evaluate vision and hearing
- Avoid abstract/noisy designs
- Avoid mirrors as can be confusing
- Use soft lighting and calm colours
- Use carpets to absorb sound
- Music

Behavioural interventions

- Identifying target symptoms
- Behavioural analysis based on analysis of antecedents, behaviours and consequences
- Set realistic goals

- Intervention
- Monitor response, e.g. Cohen-Mansfield agitation inventory
- Use reinforcement to maintain desirable behaviour
- Ensure safety throughout the process
- Avoid punishment and preserve dignity

Pharmacology

- Use only after behavioural interventions have been trialled
- 3T approach
 - Target symptoms e.g. sleep
 - Titrate from a low starting dose
 - Time-limit prescriptions
- Antipsychotics
- Low dose olanzapine or risperidone
- **RANZCP position statement** recommends oral risperidone as opposed to olanzapine
- All antipsychotics increase the risk of stroke in dementia
- Antipsychotics decrease quality of life and increase mortality and morbidity
- Donepezil, Galantamine and Rivastigmine are all effective in reducing behavioural disturbance in dementia
- Use BZD's with caution as can increase risk of falls

Management of carer's burden

- Carers assessment –aiming to identify burden, specific problems, fears and worries
- Assess additional support, coping styles, finances
- Address cultural issues and maintain cultural and ethnic sensitivity
- Psychoeducation – inform diagnosis, education about BPSD, and services and benefits that are available
- Support and advice; doctor, advocacy, carer groups
- Referral to psychologist for counselling, stress management, improve coping skills
- Liaison with psychogeriatric services and GP to provide timely care
- Other community services; meals on wheels, incontinence nurses, aids, adaptation and safety, home help and supervision
- Placement issues; respite care, day care, low- and high-level nursing homes
- Legal issues; guardianship, power of attorney, advance directives

Pharmacological options in the management of dementia

- Acetylcholinesterase inhibitors (AChE)- (*NICE Guidelines*)
- Donepezil, rivastigmine, galantamine
- Donepezil and galantamine are selective inhibitors of AChE
- Rivastigmine acts on both AChE and butyrylcholinesterase (BuChE)
- Galantamine also affects nicotinic receptors
- Donepezil can be given once daily but Rivastigmine and Galantamine must be given twice daily
- Recommended in mild to moderate dementia
- Carer's view of the condition at baseline and follow up should be sought

- Patients who continue on the drug should be reviewed every 6 months by MMSE score and global, behavioural and functional assessment
- Continue only if improvement noted
- Carers view should be sought before cessation
- Adverse effects- nausea, vomiting, insomnia and diarrhoea
- Doses- Donepezil- 5-10mg/day
Rivastigmine- start at 1.5mg B.D to max 6mg B.D
Galantamine- 12 mg B.D

Memantine

- NMDA receptor antagonist
- Blocks excitotoxic glutamate activity
- Indicated for moderate to severe dementia

Vitamin E

No conclusive evidence for effectiveness.

Ginkgo biloba

Traditional Chinese medicine made from leaves of maidenhair tree

Cognitive enhancer

May increase the risk of bleeding

Oestrogen

May delay the onset but not currently recommended

Anti-Inflammatory drugs

HMG-CoA reductase inhibitors –Statins

AntiAmyloid strategies

} No conclusive evidence

NON-PHARMACOLOGICAL TREATMENT INTERVENTIONS IN DEMENTIA

Behaviour oriented approaches

- Toilet training
- Behavioural analysis

Emotional Oriented approaches

- Reminiscence therapy
- Validation therapy
- Supportive psychotherapy
- Sensory integration
- Simulated presence therapy

Cognition oriented approaches

- Reality orientation and skills training

Stimulation oriented approaches

Art therapy Music therapy

Dance therapy Aromatherapy

DEPRESSION IN OLD AGE

Risk factors for developing depression in the elderly

- Physical illness
- Feeling lonely (not just living alone)
- Recent bereavement or other adverse event
- Hearing difficulties
- Past or family history of depression
- Dementia, early or retained insight
- Medication (steroids, major tranquilisers)
- Female gender

Differences between early and late onset depression

Clinical – in late onset depression the following are more common:

- Reduced complaint of sadness
- Hypochondriasis and somatic concerns predominant
- Poor subjective memory or a dementia like picture (cognitive deficits)
- Late onset neurotic symptoms (OCD, marked anxiety)
- Apathy and poor motivation
- Greater psychomotor retardation
- Greater incidence of completed suicide
- Less familial risk –

Anatomical

- Vascular disease (vascular depression) - Alexopoulos (1997)- damage to fronto-subcortical circuits can predispose, precipitate and perpetuate late onset depression.
- Greater neuropsychological deficits

Theories explaining the association between vascular depression and vascular disease

- Increased platelet aggregation
- Both depression and ischaemia may be secondary to atherosclerosis
- Recurrent depression across the life span may increase the risk of vascular pathology
- Damage to end arteries supplying subcortical triato-pallido-thalamo-cortical pathways may disrupt the neurotransmitter circuitry involved in mood regulation, causing or predisposing to depression

PSYCHOSIS IN OLD AGE

Characteristics of Late-Onset Schizophrenia compared to early onset schizophrenia

- Onset > 45yrs, >65yrs is very late onset schizophrenia
- Greater sensory impairment
- Social isolation
- Eccentric pre-morbid personality
- Greater likelihood of visual hallucinations
- Greater risk of developing tardive dyskinesia
- Encapsulated delusions
- Partition delusions
- Greater female preponderance
- Lesser genetic risk, i.e., family history
- No past history of schizophrenia
- Lesser formal thought disorder
- Lesser likelihood of affective blunting

CONTRIBUTORY FACTORS TO PSYCHOSIS IN OLD AGE

- Age related deterioration of frontal and temporal cortices
- Neurochemical changes of aging
- Social isolation
- Sensory deficits
- Cognitive decline
- Age related pharmacodynamic and pharmacokinetic changes
- Polypharmacy

PRINCIPLES OF PRESCRIBING FOR OLD-AGE PATIENTS

- Altered pharmacokinetics and pharmacodynamics
- Receptors become more sensitive therefore more susceptibility to side effects
- Gut motility decreases therefore decreased absorption and slower onset of action
- More body fat, less body water and less albumin, leading to higher concentration of drugs, increased duration of action for fat soluble drugs (antipsychotics) and increased levels of free drug as opposed to protein bound drug (warfarin)
- Reduced liver function may be possible
- Decline in renal function with age, exacerbated by diabetes and hypertension which is common in elderly
- Elderly individuals are usually on a multitude of medications and are therefore prone to a number of pharmacokinetic and pharmacodynamic interactions
- B-Blockers, ACEI, thiazide diuretics can cause depression
- Antidepressants are associated with SIADH in elderly – check urine and serum osmolality
- SIADH can present with worsening of depression /delirium and low sodium levels
- In elderly use drugs only when absolutely necessary

- Avoid, if possible, drugs that block alpha -1 adrenoreceptors, anticholinergics, and long acting sedative agents as these can cause confusion
- **Start low and Go Slow**